Redrawing the Industrial Map in China:
A Spatial Political Economy Perspective*

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Abstract

Chinese provinces' industrial compositions have become more similar, raising concerns about wasteful duplication. I develop a spatial political economy framework with subsidy competition. Through a revealed-preference approach, I recover governments' industrial priorities beyond resident welfare using production, trade and subsidy data. Revealed priorities become more similar across inland and coastal provinces over 2007–17. They align with national and provincial Five-Year Plans and favor industries with faster preceding local growth. Counterfactuals show that priorities partly offset provinces' limited incentives to support industries with national benefits. A constrained-optimal redesign raises national welfare and shifts priorities toward several upstream industries.

Keywords: industrial policy, provincial government, spatial equilibrium, external economies of scale, revealed priorities, Chinese economy.
JEL codes: F12, H71, O25, R12, R58.

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1 Introduction

Industrial policy is often a local decision with national consequences. Local governments compete to expand production in industries they favor, and their policies affect other provinces and sectors through trade, input-output linkages and productivity spillovers. The national consequences of this competition depend on which industries governments favor and how strongly they prioritize each of them. However, these industrial priorities are not directly observable from the data. Observed subsidies and production are equilibrium outcomes: governments choose subsidies based on their expected effects on local welfare and production, while firms adjust production in response to government policy. This paper develops a spatial political economy model and uses a revealed-preference approach to recover local governments' industrial priorities. I find that the revealed priorities become more similar across inland and coastal provinces and partly offset local governments' limited incentives to support industries with national benefits. I then construct redesigned priorities that raise national welfare, with greater weight on several upstream industries.

China provides a setting to study these questions. I document that Chinese provinces have become increasingly similar in their industrial composition. A 2012 policy proposal reported that 300 of China's 600 cities were developing solar photovoltaic (PV) industries, and more than 100 had built PV industrial bases.[1] Vehicle production is also widespread across the country: in 2016, 27 of China's 31 provincial-level regions produced at least 10,000 vehicles (Figure 1). By 2017, more than 20 provinces and cities had prioritized robotics, more than 40 robotics industrial parks had been built or were under construction, and the number of robotics firms exceeded 800.[2] These patterns raise concerns that provincial governments compete to develop the same industries, leading to wasteful duplication of production. To examine this concern and understand the role of provincial governments' objectives, I develop a framework that uses observed production and subsidies to recover these objectives and evaluate the welfare effects of their industrial priorities.

Building on quantitative trade and spatial models (Eaton and Kortum 2002, Allen and Arkolakis 2014), I study provincial subsidy competition in an economy linked by trade and intermediate inputs (Ferrari and Ossa 2023).[3] Provincial governments value both resident welfare and local sectoral value added, with priorities entering their objectives in the spirit of Grossman and Helpman (1994). I then use observed production and subsidies to recover the additional value provincial governments attach to developing particular industries. I examine

[1] See the 2012 proposal published by the Jiangsu Provincial Political Consultative Conference.

[2] See the report in *China Newsweek*, April 14, 2017.

[3] For related trade and spatial models, see also Krugman (1991), Caliendo and Parro (2015), Tombe and Zhu (2019). For a review of quantitative spatial economics, see Redding and Rossi-Hansberg (2017).

how these priorities differ across provinces, assess their relationship with development plans and industrial growth, and evaluate their welfare effects. Finally, I derive priorities that raise national welfare within each province's observed subsidy budget.

The model describes competition among provincial governments in a multi-region, multi-sector Armington economy. Production uses labor, a fixed factor and intermediate inputs, and provinces trade goods with one another (Caliendo, Parro, Rossi-Hansberg, and Sarte 2018). National external economies of scale allow expansion in an industry to raise its productivity in every province, following Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025). Each government chooses subsidies to maximize an objective that values local sectoral real value added in addition to resident welfare. The governments also engage in Nash competition, each taking other provinces' policies as given and accounting for the effects of subsidies on equilibrium outcomes such as prices and trade patterns.

Provincial industrial priorities measure the additional value a government places on local sectoral value added beyond the benefits already included in resident welfare. To understand why government priorities are different from observed subsidies, suppose one province is more productive in information and communication technology (ICT) than another. When both governments place the same priority on ICT, the productivity gap can lead subsidies to have heterogeneous effects on local production and resident welfare, leading the governments to choose different subsidy rates. A higher observed subsidy therefore need not indicate a stronger industrial priority. To recover the priorities from observed data, I calculate the marginal equilibrium responses to each province's policies and use the government's first-order conditions to infer its priorities (Ossa 2014, Adão, Becko, Costinot, and Donaldson 2026). Given the marginal responses, the first-order conditions form a linear system in the unknown weights. This revealed-preference approach separates the additional value governments attach to industrial development from the effects of subsidies on resident welfare.

The quantitative analysis covers 30 provinces[4] and 47 industries in China in 2007, 2012, and 2017. I combine production data, firms' reported cash subsidies and tax relief, and interregional input-output tables. I reconcile these data and compute the corresponding competitive equilibrium for each year. Using the governments' first-order conditions, I recover the weights provincial governments place on sectoral value added in addition to resident welfare that rationalize their observed subsidy choices. The revealed priorities show increasing alignment between inland and coastal provinces from 2007 to 2017. The ICT sector illustrates this change: the average priority inland governments place on ICT rises over time, narrowing the gap with the coast. The increasing similarity in industrial composition is therefore accompanied by greater alignment in the industries provincial governments prioritize.

[4]Excluding Tibet, Macau, Hong Kong and Taiwan due to data coverage.

National and provincial development plans help interpret the political and economic content of these priorities. Because the text data are not used to recover the priorities, they provide independent evidence about what the revealed government priorities convey. Industries identified for development in national Five-Year Plans receive higher revealed priorities. Within industries and years, a province's priority is also positively associated with designation in its own development plan, even after controlling for its subsidy rate. Within manufacturing, priorities are higher in industries with faster preceding growth in all three periods. Their conditional association with subsequent growth is positive and statistically significant in 2007, but is not statistically significant in later periods. Together, these findings connect the revealed objectives to stated policy agendas and local industrial development.

I next distinguish the effects of observed subsidies from those of the revealed priorities. In the 2012 baseline, removing all subsidies lowers national welfare by 0.23 percent. Removing sectoral priorities while allowing provincial governments to compete by choosing subsidies to maximize resident welfare lowers national welfare by 0.07 percent. Under these welfare-only objectives, governments increase total subsidy spending but reduce manufacturing's share from 46.53 to 15.39 percent. This shift speaks to the debate over whether local governments place too much emphasis on manufacturing relative to services and housing.[5] In the model, the contraction of manufacturing reduces national productivity in industries with external economies of scale, contributing to the decline in national welfare. Provincial governments account for the productivity gains only to the extent that they benefit their own residents. The revealed priorities partly offset this limited incentive to support industries with national benefits. A welfare decomposition shows that removing priorities reduces scale benefits and worsens the subsidy-distortion contribution, offsetting favorable terms-of-trade effects.

The observed allocation delivers higher welfare than either alternative, but the composition of provincial priorities can be improved. I explore this through a two-step approach. A Ramsey planner first chooses a coordinated subsidy schedule that maximizes national welfare subject to constraints on each province's real subsidy spending. I then derive redesigned priorities that give provincial governments incentives to choose this schedule under local financing. The constrained-efficient equilibrium raises national welfare by 1.28 percent in 2012, with gains in 28 of the 30 provinces and unchanged real subsidy spending in each province. Support shifts toward several upstream industries that supply intermediate inputs to other industries. Their expansion raises national industry productivity, while trade and production networks transmit the benefits to other industries in all provinces. The welfare

[5] Lu et al. (2026) document preferential land pricing for manufacturing relative to services and study its relationship with local government objectives. Henderson, Su, Zhang, and Zheng (2022) examine how local leaders' emphasis on industrial output affects the allocation of land between industrial and residential uses and the welfare of residents.

effect therefore depends on the industries being prioritized and the benefits they generate through trade and input-output linkages.

The remainder of the paper proceeds as follows. Section 2 documents the changes in industrial composition and introduces the institutional background in China. Section 3 presents the model, and Section 4 recovers and examines provincial priorities. Section 5 evaluates their effects on production and welfare. Section 6 studies the redesign of provincial priorities, and Section 7 concludes.

Related Literature. This paper contributes to four strands of literature. First, it relates to research on misallocation and regional specialization in China. Hsieh and Klenow (2009) quantify productivity losses from misallocation across firms within industries, while Brandt, Tombe, and Zhu (2013) study the misallocation of labor and capital across provinces and sectors. Focusing on the role of local governments, Young (2000) argues that governments developed high-margin industries to capture fiscal revenues and then protected those industries from interregional competition. Bai, Du, Tao, and Tong (2004) find that industries with higher past tax and profit margins are less geographically concentrated, a pattern they relate to local protection. Tombe and Zhu (2019) study how trade and migration frictions affect productivity.[6] I connect the allocation of production across provinces and industries to the objectives behind provincial subsidy choices. I also show that national productivity spillovers are important to consider in welfare analysis, as subsidies to an industry in one province can benefit other provinces.

Second, the paper contributes to quantitative analyses of industrial policy and subsidy competition. Liu (2019) shows how distortions can compound through production networks, creating a rationale for supporting upstream industries. Building on this framework, Chen, Liu, and Song (2024) study industrial policy in a multi-region economy and show that the industries a provincial government has an incentive to support may differ from those that benefit the nation. In their model, provincial governments may favor import substitution, while a national planner may favor upstream industries. Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025) estimate external economies of scale across manufacturing industries and quantify the gains from industrial policy, while Fajgelbaum and Gaubert (2020) study optimal spatial policies. Ferrari and Ossa (2023) study subsidy competition among U.S. states and compare observed policies with noncooperative and cooperative choices. I combine provincial subsidy competition with industrial priorities revealed by observed policies and national productivity spillovers. This allows me to evaluate how the objectives behind provincial subsidies affect welfare and to examine how changing those objectives can

[6] See also Lu and Tao (2009) on industrial agglomeration and Hsieh and Song (2015) on the transformation of the state sector and aggregate growth.

implement a constrained-efficient allocation within existing provincial budgets.

Third, the paper relates to China's central–local political economy. Jin, Qian, and Weingast (2005) study how provincial fiscal incentives affect local economic development. Xu (2011) and Bai, Hsieh, and Song (2020) describe the incentives and discretion of local governments. Using industrial-policy documents, Fang, Li, and Lu (2026) examine the industries targeted by governments at different levels and how policies spread across regions. They find that cities within the same province often target similar industries and relate this similarity to local protection. Wang and Yang (2025) study how local officials' incentives affect policy experimentation and national policy learning. I recover provincial industrial priorities from observed policy choices and compare them with national and provincial development plans. I use this comparison to interpret the objectives behind industrial support without imposing the stated policy agendas in the recovery. More importantly, the structural model allows me to conduct counterfactuals and evaluate the equilibrium welfare consequences of changing industrial priorities, which cannot be obtained from textual analysis alone.

Methodologically, the paper builds on research that recovers government objectives from observed policies. In public finance, the inverse-optimum approach uses policy choices to infer the social weights that rationalize existing tax and transfer systems (Bourguignon and Spadaro 2012, Hendren 2020). In trade, Ossa (2014) calibrates political-economy weights using information on noncooperative and applied tariffs. Adão, Becko, Costinot, and Donaldson (2026) use U.S. tariffs to infer the welfare weights placed on individuals across industries and regions. I apply this approach to provincial subsidy competition to recover the additional value governments place on local sectoral activity beyond resident welfare and use the revealed priorities to examine how government objectives differ across provinces and industries.

2 Provincial Similarity and Institutional Background

This section first documents the increasing similarity of industrial composition across provinces, particularly between inland and coastal provinces. It then describes China's political system, focusing on vertical accountability, horizontal competition, and the instruments provincial governments use for industrial policy. While subsidies and production contain information about provincial objectives, they are endogenous outcomes that depend on fundamentals such as productivity, trade costs and external economies of scale. Therefore, a structural model is needed to recover the objectives through a revealed-preference approach and evaluate the consequences for production and welfare.

2.1 Increasing provincial industrial similarity

Market integration creates opportunities for regions to specialize. When producers can serve customers elsewhere at lower cost, one region can devote a larger share of its resources to industries in which it has a comparative advantage, produce more in those industries and obtain other goods through trade. Lower trade costs can also encourage specialization through input-output linkages and agglomeration, as in the framework of Krugman and Venables (1996). Historical experience provides examples of this process. Regional manufacturing specialization in the United States increased from 1860 to 1914 (Kim 1995) during the railroad boom. Most EU countries also became more specialized in manufacturing from the early 1980s to the mid-1990s during a period of deepening integration (Midelfart-Knarvik, Overman, Redding, and Venables 2004).

I begin with two examples to give a sense of industrial similarity across China's provinces, before moving to the broader analysis. Vehicle production is widespread, reaching nearly all provinces in China. Figure 1 shows that 27 of China's 31 provincial-level regions produced at least 10,000 vehicles in 2016.



Figure 1: Vehicle production across China, 2016

Notes: The lowest category includes unreported production and output below 10,000 vehicles. Source: *China Statistical Yearbook 2017*.

The ICT sector illustrates how the composition of inland manufacturing has changed over

time while receiving more subsidy per unit of value added relative to the coast. The left panel of Figure 2 reports ICT's share of regional manufacturing value added. Between 2007 and 2016, ICT's value-added share increased inland while remaining relatively stable on the coast. The right panel shows that subsidies relative to ICT value added were higher inland than on the coast during the same period.

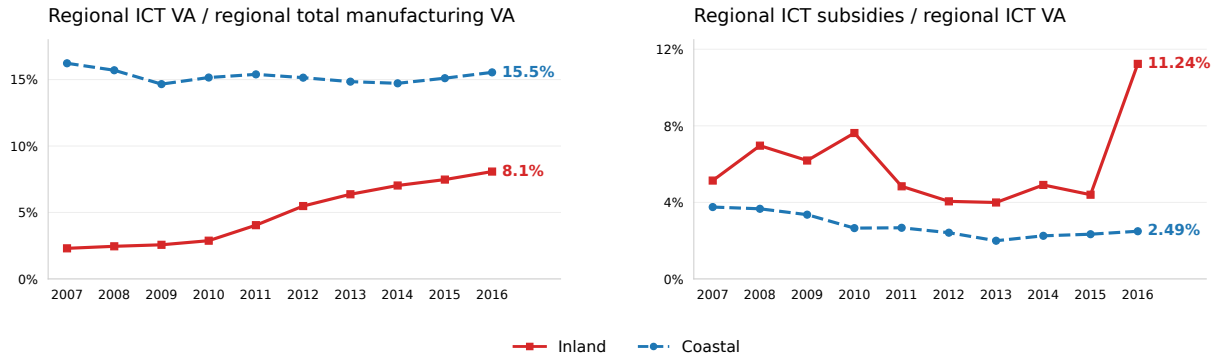


Figure 2: Regional production and fiscal support: an illustration from ICT
Notes: The left panel shows ICT's share of regional manufacturing real value added. The right panel shows subsidies and tax relief as a share of ICT value added.

I next measure industrial composition across all manufacturing sectors. For the broader pattern, I use sectoral shares of provincial manufacturing value added to capture differences in industrial composition despite differences in the overall size of manufacturing. For each province o, define

$$x_{os} = \frac{\mathrm{VA}_{os}}{\sum_j \mathrm{VA}_{oj}}, \quad \mathbf{x}_o = [x_{o1}, x_{o2}, \ldots, x_{o29}]',$$

where VA_{os} is the real value added of sector s in province o, x_{os} is the share of sector s in province o's total manufacturing value added and the vector $\mathbf{x}_o$ collects these shares for the 29 manufacturing sectors in province o. I compute three-year averages of value added to smooth out short-run fluctuations year to year.

For two share vectors, cosine similarity is defined by

$$\mathrm{Cos}(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}'\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}. \tag{1}$$

For value-added shares, the cosine lies between zero and one, with one indicating identical industrial composition. A higher cosine corresponds to a smaller angle between the two share vectors and hence more similar vectors. To illustrate how each province's industrial composition evolves, I calculate the cosine between its share vector and a fixed coastal benchmark. The benchmark is the average of the coastal provinces' manufacturing-share vectors in 2006–08. Changes in the cosine hence reflect changes in the province's own

production shares.

Figure 3 reports the regional averages. The inland mean cosine rises from 0.619 in 2006–08 to 0.688 in 2015–18. The coastal mean remains high and changes little over time. Inland provinces, on average, become more similar to the earlier coastal industrial mix, narrowing the gap with coastal provinces.

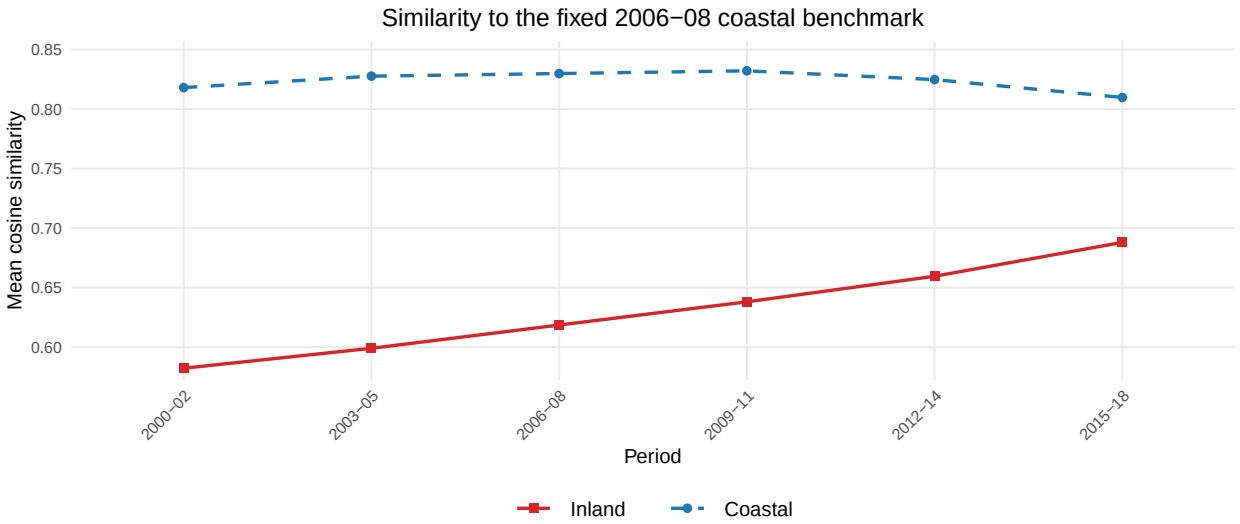


Figure 3: Industrial composition relative to a fixed coastal benchmark

This pattern also appears using other measures of industrial composition. Comparing provinces with one another in each period, the average cosine similarity across inland–coastal pairs rises from 0.521 in 2006–08 to 0.594 in 2015–18. Similarity within each region changes little over the same period: the inland-pair mean remains at 0.615, while the coastal-pair mean moves from 0.659 to 0.654.[7] Appendix Table F.1 reports these comparisons. An alternative measure, the Krugman specialization index, gives a similar account of the inland change. The mean Krugman specialization index for inland provinces, which measures their distance from the industrial composition of the rest of China, falls from 0.379 to 0.331 (Appendix Figure F.1). Using employment shares instead of value added, the increase in inland–coastal similarity is also present (Appendix Figure F.3).

These changes occurred during an expansion of China's transport network. The National Trunk Highway System added about 35,000 kilometers of high-speed roads between 1992 and 2007, connecting provincial capitals and other major cities (Faber 2014). Egger, Loumeau, and Loumeau (2023) document improvements in the extent and quality of road and rail connections between 2000 and 2013. Accounting for changes in network coverage and design

[7] An inland–coastal pair consists of one inland and one coastal province. An inland pair consists of two inland provinces, and a coastal pair consists of two coastal provinces.

speeds, Ma and Tang (2024) calculate that the median reduction in travel time across city pairs between 1994 and 2017 was 32 percent by road and 37 percent by freight rail. This evidence points to lower transport costs and better market access for firms. In standard trade and spatial models, these changes can encourage provinces to specialize in industries where they have a comparative advantage.

2.2 China's political system and provincial industrial policy

China combines centralized control over personnel with local responsibility for economic development and other policy tasks. The central leadership controls key appointments, while local governments make decisions about sectoral investments, infrastructure provision, and local economic development (Xu 2011). Provinces occupy an intermediate position in this hierarchy. Provincial leaders answer to the central authorities and oversee the cities and counties within their jurisdictions. Their policy choices therefore reflect both the demands of higher levels of government and the economic conditions they face locally.

Local economic performance can be important for officials' career prospects. Through the Chinese Communist Party's personnel hierarchy, higher authorities evaluate officials and control their appointment, promotion, and reassignment. Evidence from leaders between 1979 and 1995 shows that better economic performance was associated with a higher probability of promotion and a lower probability of termination (Li and Zhou 2005). These career incentives encourage officials to pursue local economic growth. At the same time, officials are responsible for multiple policy tasks, such as environmental regulations and social stability. Zhou (2022) describes how officials operate within these overlapping demands while retaining discretion over implementation. Provincial objectives can therefore reflect both economic performance and the tasks for which officials are held accountable.

Horizontal competition reinforces these incentives. Provincial governments compete to attract investment and expand production in particular industries. The territorial organization of government allows higher authorities to compare economic performance across jurisdictions with similar administrative responsibilities (Xu 2011). Fiscal incentives further strengthen their interest in local industrial development: under the earlier fiscal-contracting system, stronger marginal revenue-retention incentives were associated with faster development of the nonstate sector (Jin, Qian, and Weingast 2005). This competition often takes the form of selective support for firms and investment projects in the industries that local governments seek to develop (Bai, Hsieh, and Song 2020).

Provincial and lower-level governments have several instruments for pursuing these objectives. Cash subsidies and tax relief can be combined with land provision, infrastructure,

and assistance with administrative approvals. Many of these interventions are implemented by municipal governments and development-zone authorities that work directly with firms. The studies collected in Zeng (2010) describe how local governments support industrial development through roads, utilities, business services, and specialized zones. Bai, Hsieh, and Song (2020) show how political and administrative discretion can also be used to provide firm-specific arrangements for favored private businesses. Local support can consequently affect both the direct cost of production and the conditions under which a firm invests and operates. This paper uses observed output and subsidies to recover provincial industrial priorities, taking other policy instruments as given when evaluating subsidy changes. The revealed priorities therefore describe government objectives in the observed economy.

Xiong (2026) describes the government competition as an industrial tournament in which local governments want to build industries around major producers. Attracting a major producer can generate demand for local suppliers and encourage related firms to locate nearby. These potential benefits, together with fiscal and career incentives, give officials reasons to favor particular industries. Section 4.5 compares the revealed priorities with national and provincial plans as well as industrial growth. These comparisons link recovered priorities to stated policy agendas and observed industrial performance.

Officials' prior experience also affects which industries they support. Lin, Liu, Xu, and Zhang (2024) find that cities increase support for industries in which a newly appointed leader's previous city had a comparative advantage, followed by stronger export performance in those industries. Their findings connect local industrial policy to knowledge carried by officials across postings, providing a further reason for governments to favor particular industries.

Apart from the political setting, industrial policy can also address market failures and improve economic efficiency. External economies of scale provide one such rationale: when an industry's expansion raises the productivity of other firms, it creates benefits that individual producers do not fully internalize (Chipman 1970, Ethier 1982). Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025) estimate external economies of scale across manufacturing industries and quantify the gains from industrial policy. These productivity gains can give governments a welfare-based reason to support particular industries, in addition to the fiscal and career incentives described above.

The relationship between provincial similarity and external economies of scale is relevant for evaluating this possibility. Using the calibrated scale elasticities, Appendix Figure F.4 compares inland similarity to the fixed coastal benchmark within two groups of manufacturing industries, split at the median scale elasticity. Inland provinces become more similar to the fixed coastal benchmark within the group of industries with larger scale elasticities. Similarity

changes little within the other group. If expanding an industry generates productivity gains across China, several provinces' support for that industry may have benefits beyond their own borders. On the other hand, such support also uses fiscal resources and draws labor and inputs away from other activities. Its welfare consequences depend on both sides of this trade-off.

Neither observed subsidies nor output directly identify provincial priorities because they are endogenous equilibrium outcomes. Firms respond to subsidies, productivity, market access, and input costs. Governments choose subsidies to maximize their objectives, taking into account how their policy choices affect the competitive equilibrium. Two provinces with similar objectives can therefore choose different subsidy rates if the marginal effects of subsidies on production and resident welfare differ. Conversely, similar rates can reflect different objectives and industrial priorities. Production shares also reflect both economic fundamentals and policy support. Recovering provincial priorities requires accounting for these general equilibrium effects, including those transmitted across provinces through trade, national productivity spillovers and input-output linkages. Section 3 develops a structural model for the revealed-preference approach.

3 Model

Based on the observed facts and understanding of the institutional setting, this section develops a spatial political economy model to capture provincial competition and government objectives. The model builds on a multi-region multi-sector Armington trade model with external economies of scale, and introduces government objectives. I first present the competitive equilibrium for households and firms, taking subsidy rates as given. I then discuss how provincial governments choose subsidies to maximize their objectives, evaluated at the competitive equilibrium. The recovery procedure uses governments' first-order conditions to derive the additional value provincial officials attach to local sectoral activity.

The economy contains provinces $o, r \in \mathcal{R}$ and industries $s, j \in \mathcal{S}$. Goods are traded among provinces and differentiated by their province of origin within each industry, following Armington (1969). Production uses labor, a fixed factor that can be interpreted as capital, and intermediate inputs, with input-output linkages across industries as in Caliendo and Parro (2015). National external economies of scale follow the composite-input formulation of Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025). For simplicity, I assume that provincial populations and fixed-factor endowments are immobile across provinces, while

both factors can move between industries within a province.[8]

3.1 Production technology and firm behavior

The production technology in province o and industry s is given by

$$q_{os} = \bar{A}_{os}\Psi_s F_{os}, \tag{2}$$

where q_{os} denotes output in province o and industry s, $\bar{A}_{os}$ is fundamental productivity, Ψ_s is the national productivity multiplier, and F_{os} is the constant-returns input composite defined by

$$F_{os} = \left(K_{os}^{\alpha_s} L_{os}^{1-\alpha_s}\right)^{1-\gamma_s} \prod_j M_{os,j}^{\gamma_{sj}}.$$

Here L_{os} is labor, K_{os} is the fixed factor, and $M_{os,j}$ is the quantity of intermediate input j. The coefficient γ_{sj} is the share of industry s's total input expenditure allocated to inputs from industry j. The total intermediate-input share is $\gamma_s = \sum_j \gamma_{sj}$. The value-added share is $1 - \gamma_s$ and α_s is the fixed factor's share of primary-factor payments.

National productivity depends on the physical input composite used in industry s across all provinces:

$$\Psi_s = Z_s^{\phi_s} = \left(\sum_{o \in \mathcal{R}} F_{os}\right)^{\phi_s}, \tag{3}$$

where $\phi_s \geq 0$ is the elasticity of sectoral productivity with respect to the national input composite and Z_s is the total physical input composite employed in industry s across all provinces. A larger ϕ_s implies stronger productivity spillovers, while $\phi_s = 0$ implies no scale effects. For industries with positive scale elasticities, an increase in Z_s raises productivity in industry s in every province.

The national scale term captures productivity spillovers across provincial borders in reduced form. For example, Cai, Caliendo, Parro, and Xiang (2025) find that migration contributes to knowledge accumulation in Chinese provinces, particularly when migrants come from more productive provinces. Productivity benefits also travel through firms operating in multiple locations: Giroud, Lenzu, Maingi, and Mueller (2024) document gains at distant plants connected to locations experiencing positive productivity spillovers.[9] These findings motivate allowing industrial expansion in one province to raise productivity elsewhere. The

[8] The model can be extended to allow for labor mobility. A calibrated labor-mobility extension preserves the recovered pattern of priorities and the increase in inland–coastal alignment.

[9] Province-level evidence also links Chinese industrial productivity to knowledge capital accumulated in other provinces (Scherngell, Borowiecki, and Hu 2014). Peri (2005) documents interregional knowledge flows and their contribution to innovation.

national input pool represents this transmission, while industry-specific scale elasticities govern its strength. Appendix G.5 considers productivity spillovers confined to individual provinces.

Firms take the national productivity multiplier Ψ_s as given.[10] For any province o, let p_{os} denote the factory-gate output price, r_o the fixed-factor rent, w_{os} the sector-specific wage, and P_{oj} the price of the industry-j composite in province o. A subsidy rate $\tau_{os} < 1$ reimburses the fraction τ_{os} of the firm's expenditure on all inputs. While I allow for negative subsidy rates which are effectively taxes, I will use the term 'subsidy' for simplicity. Firms choose output and inputs to maximize profits:

$$\begin{aligned} \max_{\substack{q_{os},K_{os},L_{os}\geq 0 \\ M_{os,j}\geq 0\ \forall j}} \quad & p_{os}q_{os} - (1-\tau_{os})\left(r_o K_{os} + w_{os}L_{os} + \sum_j P_{oj}M_{os,j}\right) \\ \text{subject to} \quad & q_{os} \leq \bar{A}_{os}\Psi_s \left(K_{os}^{\alpha_s} L_{os}^{1-\alpha_s}\right)^{1-\gamma_s} \prod_j M_{os,j}^{\gamma_{sj}}. \end{aligned} \tag{4}$$

Because the reimbursement applies to all inputs at the same rate, it leaves the cost-minimizing input mix unchanged at given input prices. Cost minimization yields the gross cost of one unit of the input composite:

$$u_{os} = C_s \left(r_o^{\alpha_s} w_{os}^{1-\alpha_s}\right)^{1-\gamma_s} \prod_j P_{oj}^{\gamma_{sj}}, \tag{5}$$

where C_s is a constant that depends on the cost shares.[11]

Let B_{os} denote the input bill before reimbursement, R_{os} sales revenue, and S_{os} subsidy expenditure. Constant returns to scale and perfect competition imply zero profits for active producers, so

$$B_{os} = u_{os}F_{os}, \qquad R_{os} = (1-\tau_{os})B_{os}, \qquad S_{os} = \tau_{os}B_{os}. \tag{6}$$

For an active producer, the factory-gate price therefore equals marginal cost net of the subsidy:

$$p_{os} = \frac{(1-\tau_{os})u_{os}}{\bar{A}_{os}\Psi_s}. \tag{7}$$

[10] This formulation can be interpreted as a continuum of perfectly competitive, atomistic firms producing the same good within each province–industry. Each firm's contribution to national productivity is negligible. Identical technologies and input prices imply common input proportions. With constant returns to scale, the production function also describes province–industry aggregates.

[11] Specifically,

$$C_s = [\alpha_s(1-\gamma_s)]^{-\alpha_s(1-\gamma_s)}[(1-\alpha_s)(1-\gamma_s)]^{-(1-\alpha_s)(1-\gamma_s)} \prod_{j:\gamma_{sj}>0} \gamma_{sj}^{-\gamma_{sj}}.$$

The subsidy covers a share τ_{os} of the gross input bill, and producers pay $(1-\tau_{os})B_{os}$. Workers, fixed-factor owners, and intermediate suppliers together receive B_{os}, which is financed by sales revenue and the subsidy. Cost minimization implies factor payments

$$w_{os}L_{os} = (1-\alpha_s)(1-\gamma_s)B_{os}, \qquad r_oK_{os} = \alpha_s(1-\gamma_s)B_{os}. \tag{8}$$

3.2 Worker allocation and consumption

Province o has $\bar{L}_o$ workers with heterogeneous sectoral skills. Let ϵ_{hs} denote worker h's productivity in industry s, measured in efficiency units of labor. If employed in that industry, the worker earns $w_{os}\epsilon_{hs}$. A worker h draws independent sectoral skills with Fréchet distribution $\Pr(\epsilon_{hs} \le e) = \exp(-e^{-\zeta})$ for $e > 0$, where $\zeta > 1$. Taking sectoral wages as given, the worker chooses industry s_h^* to maximize earnings:

$$s_h^* \in \arg\max_s \; w_{os}\epsilon_{hs}. \tag{9}$$

Aggregating these choices gives the employment shares, mean labor earnings, and labor supplied to each sector:

$$\lambda_{os} = \frac{w_{os}^{\zeta}}{\sum_j w_{oj}^{\zeta}}, \qquad y_o = \Gamma(1-1/\zeta)\left(\sum_j w_{oj}^{\zeta}\right)^{1/\zeta}, \qquad L_{os} = \frac{\lambda_{os}\bar{L}_o y_o}{w_{os}}. \tag{10}$$

Here λ_{os} is the employment share and y_o is mean labor earnings. $\Gamma(\cdot)$ denotes the gamma function. The factor $\Gamma(1-1/\zeta)$ arises from averaging maximized earnings under the Fréchet skill distribution, with $\zeta > 1$ ensuring that mean earnings are finite. As an industry draws a larger share of workers, it attracts workers with lower relative productivity in that occupation and must offer a higher efficiency wage relative to other industries. The parameter ζ governs the dispersion of sectoral skills. A larger ζ implies less skill dispersion and makes relative employment shares more responsive to relative wages.

Residents earn labor income and own the provincial fixed-factor endowment $\bar{K}_o$, which has a common rent r_o across industries. [12] The government finances net subsidy spending through a lump-sum tax $T_o = \sum_s \tau_{os}B_{os}$. I abstract from interprovincial fiscal transfers.[13]

To account for trade imbalances among provinces, let D_o denote the net inflow to province o. A positive D_o finances a trade deficit, while a negative value corresponds to a trade surplus. Since trade takes place among domestic provinces, $\sum_{o\in\mathcal{R}} D_o = 0$. Household disposable

[12] Equivalently, the government can own the fixed factor and transfer its rental income to residents net of subsidy spending. If spending exceeds rental income, the rest is still financed by a lump sum tax.

[13] This is different from assuming balanced trade. Each province finances its own subsidies and nominal trade imbalances enter household resources through D_o.

income is

$$Y_o = y_o \bar{L}_o + r_o \bar{K}_o - T_o + D_o. \tag{11}$$

Final and intermediate users in destination r purchase the sectoral composite in sector s. Goods from different provinces are imperfect substitutes, and the sectoral composite is a CES aggregator of the delivered quantities:

$$Q_{rs} = \left(\sum_{o \in \mathcal{R}} x_{ro,s}^{(\theta_s)/\sigma_s} \right)^{\sigma_s/\theta_s}, \tag{12}$$

where $x_{ro,s}$ is the quantity delivered from province o to province r. The parameter $\sigma_s > 1$ is the elasticity of substitution between goods from different provinces within industry s, and $\theta_s \equiv \sigma_s - 1 > 0$ is the trade elasticity. A larger σ_s makes buyers more willing to substitute across provincial suppliers when relative prices change. Standard iceberg-trade cost requires $\kappa_{ro}^s \geq 1$ units at origin per unit delivered at destination, with $\kappa_{oo}^s = 1$.

Let C_{os} denote province-o household consumption of the industry-s composite. Given sectoral prices and disposable income, the representative household chooses consumption to maximize Cobb–Douglas utility:

$$\max_{\{C_{os} \geq 0\}_s} \quad \prod_{s:\beta_s > 0} C_{os}^{\beta_s} \qquad \text{subject to} \qquad \sum_s P_{os} C_{os} \leq Y_o, \tag{13}$$

where $\beta_s \geq 0$ is the household expenditure share on industry s, common across provinces, and $\sum_s \beta_s = 1$. The solution satisfies $P_{os} C_{os} = \beta_s Y_o$, so the household allocates a constant share of disposable income to each industry. Its consumption price index and indirect utility are

$$P_o^c = \prod_{s:\beta_s > 0} (P_{os}/\beta_s)^{\beta_s}, \qquad W_o = \frac{Y_o}{P_o^c}. \tag{14}$$

3.3 Competitive equilibrium

For a given subsidy profile, the competitive equilibrium is characterized as follows. Cost minimization over the origins in the sectoral composite yields the price index and expenditure shares

$$P_{rs} = \left[\sum_{o \in \mathcal{R}} (\kappa_{ro}^s p_{os})^{-\theta_s} \right]^{-1/\theta_s}, \qquad \pi_{ro,s} = \frac{(\kappa_{ro}^s p_{os})^{-\theta_s}}{\sum_{m \in \mathcal{R}} (\kappa_{rm}^s p_{ms})^{-\theta_s}}. \tag{15}$$

Here P_{rs} is the price of one unit of the industry-s composite in destination r, faced by both households and intermediate-input users. $\pi_{ro,s}$ is the share of destination r's expenditure on industry s supplied by province o. These trade shares depend on the equilibrium delivered prices of all available producers.

Total expenditure on industry s in destination r consists of household consumption and intermediate purchases. Goods-market clearing equates each origin's sales revenue with expenditure on its goods across destinations:

$$E_{rs} = \beta_s Y_r + \sum_j \gamma_{js} B_{rj}, \tag{16}$$

$$R_{os} = \sum_{r \in \mathcal{R}} \pi_{ro,s} E_{rs}. \tag{17}$$

Labor demanded by firms must equal the occupational supply in equation (10). The fixed-factor market clears when

$$\sum_s K_{os} = \bar{K}_o. \tag{18}$$

Together with the factor payments in equation (8), these conditions imply $y_o \bar{L}_o + r_o \bar{K}_o = \sum_s (1 - \gamma_s) B_{os}$. After financing subsidy expenditure and accounting for any trade imbalances, disposable income for consumers in province o can therefore be written as

$$Y_o = \sum_s (1 - \gamma_s - \tau_{os}) B_{os} + D_o. \tag{19}$$

These accounts imply $\sum_s (E_{os} - R_{os}) = D_o$. For a reference province $o^* \in \mathcal{R}$, prices are normalized by $P^c_{o^*} = 1$, with D_o expressed in the same numeraire.

Definition 1 (Competitive equilibrium)**.** Given the model's primitives, a subsidy profile $\boldsymbol{\tau} = (\boldsymbol{\tau}_o)_{o \in \mathcal{R}}$, and trade imbalances $\{D_o\}_{o \in \mathcal{R}}$ satisfying $\sum_o D_o = 0$, a competitive equilibrium consists of prices, production and input choices, consumption and trade, labor sectoral choices, and taxes such that:

(i) Firms solve the profit-maximization problem in equation (4), taking prices and national productivity multipliers as given.

(ii) Workers choose sectors following equation (9), and households solve the consumption problem in equation (13). Final and intermediate users minimize the cost of their sectoral composites.

(iii) Goods markets clear according to equations (16) and (17). Firms' labor demands equal the occupational supplies in equation (10), and the fixed-factor market clears according to equation (18).

(iv) Each provincial government balances its budget, $T_o = \sum_s \tau_{os} B_{os}$, and household disposable income satisfies equation (11). National input pools and productivity multipliers satisfy equation (3).

Appendix B gives the detailed equilibrium conditions and their exact-change representation.

3.4 Welfare effects of subsidies

Changes in subsidies affect resident welfare through trading prices, national productivity, and the allocation of production inputs. Holding fundamentals, trade costs, and nominal provincial trade imbalances fixed, the equilibrium welfare change satisfies

$$\begin{aligned} d\log W_o = & \underbrace{\frac{1}{Y_o}\sum_s \left(R_{os}\, d\log p_{os} - E_{os}\, d\log P_{os}\right)}_{\text{Terms of trade}} \\ & + \underbrace{\frac{1}{Y_o}\sum_s R_{os}\phi_s\, d\log Z_s}_{\text{Scale externalities}} \\ & + \underbrace{\left(-\frac{1}{Y_o}\sum_s S_{os}\, d\log F_{os}\right)}_{\text{Subsidy distortions}}. \end{aligned} \tag{20}$$

The standard terms-of-trade term measures changes in the prices at which a province sells its output and purchases goods for consumption and intermediate use. The scale-externality term values productivity changes generated by changes in the national industry input pools, Z_s. The subsidy-distortion term captures the welfare effects of changing physical composite input use, $F_{os} = B_{os}/u_{os}$, in the presence of subsidy wedges. For a subsidized activity, an increase in physical input use contributes negatively through this term, reflecting the gap between private input costs and resource costs. Scale externalities and subsidy distortions together give the net production-efficiency contribution. All terms are evaluated at the current equilibrium.

3.5 Government objectives and Nash equilibrium

The competitive equilibrium describes the economy under a given subsidy schedule chosen by provincial governments. The institutional setting in Section 2.2 combines accountability to national authorities with competition for local economic activity. Provincial officials value certain industries because of GDP and promotion incentives, assigned industrial goals, or expectations about future growth. These motives can lead them to support sectoral expansion beyond the amount justified by the welfare gains received by their residents. I represent these considerations with an additional valuation of sectoral activity, in the spirit of the flexible political objectives in Grossman and Helpman (1994) and Ossa (2014). Government

o chooses its subsidy vector $\boldsymbol{\tau}_o = (\tau_{os})_{s \in \mathcal{S}}$ to maximize

$$g_o(\boldsymbol{\tau}; \boldsymbol{\vartheta}_o) = \log W_o(\boldsymbol{\tau}) + \sum_s \vartheta_{os} \log V_{os}(\boldsymbol{\tau}), \tag{21}$$

where $V_{os} = (w_{os} L_{os} + r_o K_{os})/P_o^c$ is the real value added of sector s in province o.

The first term is the log of resident welfare, while the second term is a weighted sum of log sectoral activities. The weights ϑ_{os} are the unobserved priorities that summarize the value officials attach to local sectoral activity in addition to the resident welfare.

The government holds other provinces' subsidies fixed, anticipates the competitive-equilibrium response to each candidate policy, and finances its resulting subsidy bill. Each local government takes into account how its policies affect factor prices, trade, consumer prices, and national productivity when evaluating their effects on resident welfare and local sectoral activity.

I provide two ways to interpret these priorities.

Government Indifference Curve: Similar to the concept of an indifference curve in consumer theory, a priority ϑ_{os} represents the rate at which the government is willing to trade off welfare for a proportional increase in sectoral real value added. Taking the total differential of g_o, with priorities held fixed, gives

$$dg_o = d \log W_o + \sum_j \vartheta_{oj} \, d \log V_{oj}.$$

Holding the objective fixed, $dg_o = 0$, and other sectoral activities fixed, $d \log V_{oj} = 0$ for $j \neq s$, yields

$$d \log W_o = -\vartheta_{os} \, d \log V_{os}.$$

For example, $\vartheta_{os} = 0.2$ means the government is willing to reduce resident welfare by approximately 0.2% to increase the real value added of industry s by 1%, ceteris paribus.

Hypothetical resident: Consider a hypothetical consumer who values both consumption and local sectoral value added. The combined weight this resident places on consumption and local value added in industry s is

$$\widetilde{\beta}_{os} = \frac{\beta_s + \vartheta_{os}}{1 + \sum_j \vartheta_{oj}}. \tag{22}$$

When all priorities are zero, these weights coincide with household consumption shares. When subsidies change, the hypothetical resident considers their effects on consumption and local value added across all industries.[14]

[14] This interpretation applies when $\beta_s + \vartheta_{os} \geq 0$ for every industry and $1 + \sum_j \vartheta_{oj} > 0$.

The governments maximize their objectives by choosing subsidies, taking the other provinces' subsidies as given. A Nash equilibrium is a subsidy profile and its associated competitive equilibrium such that no province can improve its objective by unilaterally changing its own subsidies. The following definition formalizes this concept.

Definition 2 (Nash equilibrium)**.** Given provincial priorities $\{\boldsymbol{\vartheta}_o\}_{o\in\mathcal{R}}$, a Nash equilibrium is a subsidy profile $\boldsymbol{\tau}^*$ and the competitive equilibrium induced by it such that

$$g_o(\boldsymbol{\tau}_o^*, \boldsymbol{\tau}_{-o}^*; \boldsymbol{\vartheta}_o) \geq g_o(\boldsymbol{\tau}_o', \boldsymbol{\tau}_{-o}^*; \boldsymbol{\vartheta}_o) \tag{23}$$

for every province $o \in \mathcal{R}$ and every feasible unilateral deviation $\boldsymbol{\tau}_o' = (\tau_{os}')_{s\in\mathcal{S}}$. Here $\boldsymbol{\tau}_{-o}^*$ denotes the subsidy vectors of all other provinces. A deviation is feasible if $\tau_{os}' < 1$ in every industry and the resulting competitive equilibrium yields a well-defined government objective. Both payoffs are evaluated at the competitive equilibrium induced by the corresponding subsidy profile.

3.6 Recovering priorities from subsidies and output

The government's optimality conditions link its unobserved priorities to the observed subsidy schedule. At an interior choice, a marginal increase in any subsidy must balance the resulting welfare change against the weighted changes in sectoral activities the government values. I evaluate equilibrium responses at observed policies and ask which priority vector makes these tradeoffs consistent with provincial choices. For each industry $k \in \mathcal{S}$, define

$$(a_o)_k = \frac{\partial \log W_o}{\partial \tau_{ok}}, \qquad (\mathcal{B}_o)_{ks} = \frac{\partial \log V_{os}}{\partial \tau_{ok}}. \tag{24}$$

Each element $(a_o)_k$ measures the change in log resident welfare from a marginal increase in the subsidy to industry k. The corresponding row of $\mathcal{B}_o$ records the changes in log real value added across all local industries following a marginal increase in the subsidy to industry k. The diagonal entry captures the response of the subsidized industry, while the other entries capture the responses of other industries. These are the reasons subsidy rates need not rank priorities: a higher subsidy can reflect differences in its effects on resident welfare and sectoral activity, rather than a higher priority alone.

The FOC implies that $a_o + \mathcal{B}_o\vartheta_o = 0$. With a nonsingular square response matrix, these conditions uniquely recover $\vartheta_o = -\mathcal{B}_o^{-1}a_o$. The effects of a subsidy on resident welfare and sectoral activity depend on the province's productivity, production structure, and trade connections. The revealed priorities describe the additional valuations required to balance those marginal responses at the observed policies. I recover priorities for each province and

year. Section 4.5 interprets them through their relationships with national and provincial development plans and industrial growth.

The marginal welfare return has three components:

$$(a_o)_k = \underbrace{\sum_s \frac{(1-\gamma_s-\tau_{os})B_{os}}{Y_o}\frac{\partial \log B_{os}}{\partial \tau_{ok}}}_{\text{Income response}} - \underbrace{\frac{B_{ok}}{Y_o}}_{\text{Direct subsidy cost}} - \underbrace{\sum_s \beta_s \frac{\partial \log P_{os}}{\partial \tau_{ok}}}_{\text{Consumer price response}}. \tag{25}$$

The first term measures how changes in sectoral input bills affect disposable income. An additional unit of B_{os} generates $1-\gamma_s$ units of factor income and requires τ_{os} units of subsidy spending, leaving a net contribution of $1-\gamma_s-\tau_{os}$. The second term is the direct cost of increasing the subsidy rate in industry k, holding its input bill fixed, relative to disposable income. The third term captures changes in the consumption price index, with lower prices raising resident welfare.

Equation (25) expresses the marginal welfare return through income, subsidy financing, and consumer prices. Equivalently, equation (20) separates this return into terms-of-trade, scale-externality, and subsidy-distortion effects. Provincial governments value each channel through its effect on their own residents, while sectoral priorities add the valuation of local activity in equation (21).

Together with the first-order conditions, a negative-definite own-policy Hessian is sufficient for strict local optimality. At the revealed priorities, the Hessian of province o's objective with respect to its subsidy vector is

$$H_o = D^2_{\tau_o} \log W_o + \sum_s \vartheta_{os} D^2_{\tau_o} \log V_{os}. \tag{26}$$

4 Recovering Provincial Priorities

This section takes the model to the data to recover provincial industrial priorities. I first introduce the data and explain how observed production, subsidies, and trade enter the recovery. I then show how the revealed priorities differ across provinces and change over time, and how they relate to national and provincial development plans and sectoral growth. These comparisons help interpret the objectives behind revealed priorities.

4.1 Data

The quantitative analysis covers 30 provinces and 47 industries in 2007, 2012, and 2017. Production data come from the *China Industry Statistical Yearbook* and provincial input-output tables. The China Tax Survey provides data on cash subsidies and tax relief, which I

combine into a measure of subsidy for each province-industry pair. Each benchmark combines production and policy observations from nearby years to smooth out short-run fluctuations. Appendix A describes the coverage, timing, and construction of these measures.

The interregional input-output tables compiled by Li et al. (2023) provide trade flows between provinces and with the rest of the world. These flows describe where each province's industries buy their inputs and sell their output. In the baseline, I calibrate the cost shares γ_s and γ_{sj} and consumption shares β_s using the 2007 national input-output table and hold them fixed across periods. Provincial production, subsidies, and trade vary across periods.

I use development plans and manufacturing growth to interpret the political and economic content of the revealed priorities. The Eleventh, Twelfth, and Thirteenth national Five-Year Plans identify industries for development in the three benchmark periods (National People's Congress 2006, 2011, 2016). The CNRDS Industrial Policy Research Database records industries designated for support in provincial plans (Chinese Research Data Services 2026). I use annual manufacturing data to measure industrial growth before and after each benchmark. Neither the textual data nor the data on industrial growth enter the recovery. Their construction and the industry concordances are described in Appendix A.

4.2 Accounting for productivity differences

Before recovering provincial priorities, I briefly explain how the analysis accounts for productivity differences across provinces and over time. In the model, productivity differences enter the recovery through benchmark production accounts and trade shares, together with the calibrated parameters. To document how inland productivity has changed relative to the coast, I also calculate relative productivity through a level inversion, following the quantitative spatial literature (Redding and Rossi-Hansberg 2017). I assume that trade costs share a common province-pair component across industries, while allowing producer prices to differ across provinces and industries (Fan 2019). Observed employment and fixed assets inform factor costs. I recover productivity up to an industry-year normalization and compare inland productivity with that of coastal producers. Appendix H gives details about the procedure.

Table 1 reports estimates of inland productivity catch-up relative to coastal producers. Using the same sample of inland province-industry pairs across all three years, average relative productivity in manufacturing rises from 0.688 in 2007 to 0.802 in 2012 and 0.917 in 2017, showing inland productivity moved closer to the coastal level. The ICT series rises from 0.479 to 0.635 and 0.841, an increase of approximately 75 percent over the ten years. These auxiliary productivity estimates and the assumptions used to derive them do not enter the priority recovery.

Table 1: Conditional inland productivity relative to coastal producers

	2007	2012	2017
Manufacturing	0.688	0.802	0.917
ICT	0.479	0.635	0.841

Notes: Geometric means across the same 545 inland manufacturing province–industry cells, including 19 ICT cells, relative to coastal producers active at every benchmark.

4.3 Implementing the recovery

Because some public services are provided directly by the government in China, I hold policies in four public-service industries[15] at their observed values and allow provincial governments to choose subsidies in the remaining 43 industries. All 47 industries remain in the model, so changes in the public-service industries also affect the returns to subsidies through production and demand linkages.

The observed subsidy measure is reported relative to output, while the model expresses subsidies as a share of the gross input bill. Let s_{os} denote reported subsidies divided by reported output. The subsidy rate and gross input bill in the model are then given by:

$$\tau_{os} = \frac{s_{os}}{1 + s_{os}}, \qquad \frac{S_{os}}{R_{os}} = s_{os}, \qquad B_{os} = \frac{R_{os}}{1 - \tau_{os}}. \tag{27}$$

Calibrated Cobb–Douglas cost shares determine the payments to labor, the fixed factor, and intermediate input producers. I reconcile the observed trade profiles with the resulting production and expenditure accounts while preserving observed output and the provincial net-import ratios measured from the interregional input-output tables. Appendix B describes the accounting conventions and trade reconciliation.

China trades extensively with the rest of the world. I therefore extend the model to include imports and exports for each province. The rest of the world supplies imports at fixed prices and allocates fixed expenditure on Chinese goods in each industry across provincial producers through the same CES demand system. Export shares and quantities respond to provincial policies, while national productivity spillovers continue to depend only on production inputs employed in China. Appendix B.3 presents this extension.

The elasticities determine how production, trade, and labor allocation respond to subsidies. I calibrate trade elasticities to Giri, Yi, and Yilmazkuday (2021) and national scale elasticities to Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025). The scale estimates used

[15]These industries are (i) water, environment and public facilities, (ii) education, (iii) health and social work, and (iv) public administration and social security.

in the calibration cover manufacturing industries. I set scale elasticities to zero in other industries in the main specification. The Roy elasticity parameter is $\zeta \simeq 1.39$, based on an auxiliary calibration using employment and earnings. These parameters are held fixed across periods. Appendix B.5 reports the values, industry mapping, and occupational calibration.

I use the reconciled production and trade data, observed subsidies, and calibrated parameters to calculate the marginal welfare and activity responses. I then solve the governments' first-order conditions for their priorities. The response matrices have full rank in all 90 province-years, yielding unique provincial priority vectors consistent with the first-order conditions at the observed allocation. At these priorities, the own-policy Hessians are negative definite in all 90 province-years. The revealed priorities therefore support the observed subsidy profile as a strict local Nash equilibrium in each year. Solving the Nash equilibrium from different starting points in each year reproduces the observed policies. This agreement provides numerical support for the observed subsidy profile as the unique Nash equilibrium under the recovered priorities. Appendix C presents the derivation, and Appendix C.4 reports the recovery diagnostics.

4.4 Similarity of provincial priorities

I first ask whether the increasing similarity in industrial composition documented in Section 2 is accompanied by increasing alignment of provincial priorities. Figure 4 reports pairwise cosine similarity in revealed priorities across provinces, calculated using the 43 policy sectors and equation (1). The mean inland–coastal cosine rises from 0.27 in 2007 to 0.36 in 2012 and 0.40 in 2017. Similarity increases in 66 percent of inland–coastal pairs between 2007 and 2012, showing that the increase in the average is shared by most pairs. Most of the increase occurs by 2012, with further alignment by 2017. Inland and coastal provinces have therefore become more similar in both their industrial composition and the industries their governments prioritize.

Within each region, provincial priorities follow a different pattern. Similarity among coastal provinces falls slightly by 2012 and returns to its initial level by 2017, while similarity among inland provinces falls from a higher initial level. Rising inland–coastal alignment therefore coexists with substantial heterogeneity across provinces. The manufacturing-only inland–coastal series also rises. Appendix F reports both series.

Returning to the ICT example in Section 2, I compare its priorities across regions and over time in Table 2. Its mean inland priority rises from 0.0022 in 2007 to 0.0079 in 2012 and 0.0174 in 2017, and its rank among inland sector means improves from thirteenth to fifth to third. The coastal mean falls from 0.0606 to 0.0372 over the same period, but remains

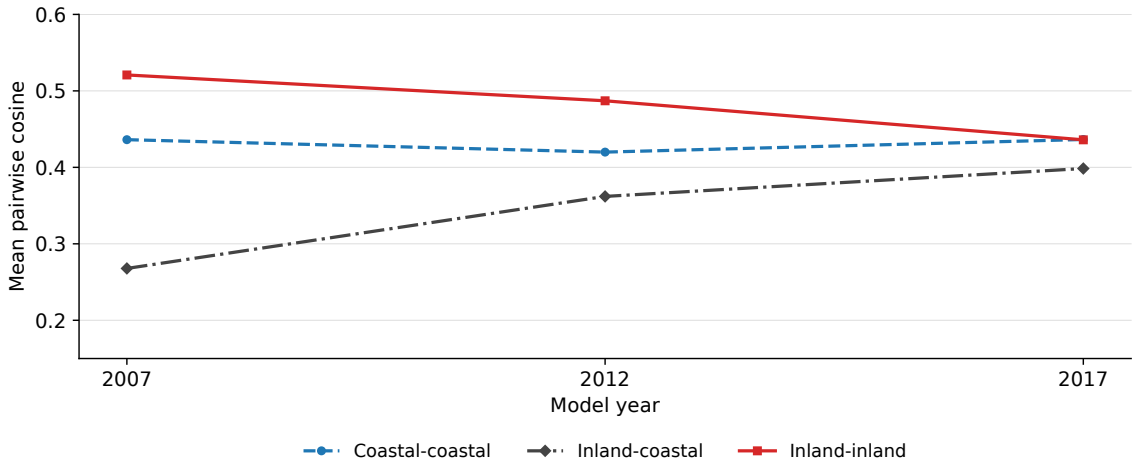


Figure 4: Similarity of provincial priorities across the 43 policy sectors

above the inland mean. The regional gap in ICT priorities narrows through both higher inland priority and a lower coastal average. The expansion of ICT in inland manufacturing is therefore accompanied by increases in both inland relative productivity, documented in Section 4.2, and the priority inland governments place on the industry.

Table 2: An illustration of changing regional priorities: ICT

Year	Coastal mean priority	Inland mean priority	Inland rank
2007	0.0606	0.0022	13
2012	0.0395	0.0079	5
2017	0.0372	0.0174	3

Notes: Means weight active provinces equally. Ranks compare inland mean priorities across the 43 policy sectors.

4.5 Priorities and political incentives

The revealed priorities can reflect several motives for provincial industrial support. Political-economy models such as Grossman and Helpman (1994) allow policy to depend on sectoral concerns as well as aggregate welfare. In China, career incentives may lead officials to place greater weight on industries prioritized by the central government or those that contribute to local economic performance. Xiong (2026) describes how central direction and local competition interact in an industrial tournament.

To assess how the revealed priorities relate to political incentives, I compare them with development plans and industrial growth. National Five-Year Plans identify industries the central government aims to develop, while provincial plans record which industries

individual governments designate for support. The comparison with prior growth asks whether governments place higher priorities on industries that have grown faster locally. I also relate priorities to subsequent growth, accounting for past performance and observed subsidies. These comparisons connect revealed priorities to political incentives and industrial performance.

National five-year plans. I use national plans to assess whether provincial priorities align with national development objectives. I match the Eleventh, Twelfth, and Thirteenth national Five-Year Plans to the 2007, 2012, and 2017 economies. A national agenda indicator equals one when an activity identified for development in the plan maps to the corresponding sector. I first regress revealed priorities, standardized within each benchmark year and set of industries, on the national agenda indicator, including province fixed effects. The coefficient measures the priority difference between industries identified for development in the national plan and other industries.

Industries identified for development in national plans receive higher revealed priorities in all three economies (Table 3). Within manufacturing, the estimated difference is 0.36 standard deviations in 2007, 0.41 in 2012, and 0.45 in 2017. Across all 43 policy sectors, the corresponding differences are 0.33, 0.42, and 0.39. The manufacturing point estimates suggest a modest increase in alignment with national plans over time. For all policy sectors, the association is also positive. These results are consistent with provincial governments placing greater weight on industries that carry the central agenda, as described in Fang, Li, and Lu (2026), Wang and Yang (2025) and Xiong (2026).

Table 3: National Five-Year Plans and revealed priorities

Revealed priority (standard deviations)	2007	2012	2017
Panel A: Manufacturing sectors			
National FYP support	0.36**	0.41***	0.45**
	(0.15)	(0.16)	(0.18)
Observations	868	865	866
Panel B: All 43 policy sectors			
National FYP support	0.33***	0.42***	0.39***
	(0.10)	(0.11)	(0.11)
Observations	1,288	1,285	1,286
Province fixed effects	Yes	Yes	Yes
Sector-clustered standard errors	Yes	Yes	Yes

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Provincial five-year plans. Provincial plans list the industries each government intends to support. These choices can reflect both national development objectives and local economic

conditions. I compare the industries named in these plans with the revealed priorities to see whether they are related, even after accounting for observed subsidy rates.

The CNRDS Industrial Policy Research Database records industries designated for local support and the associated text in provincial Five-Year Plans (Chinese Research Data Services 2026). I match the activities named in the database to the model sectors and define $D_{os,t} = 1$ when the database records sector s as a key industry for support in province o's plan for year t. Pooling the three benchmark years, I regress the provincial support indicator on revealed priorities and observed subsidy rates:

$$100 D_{os,t} = b_\vartheta z_t(\vartheta_{os,t}) + b_\tau z_t(\tau_{os,t}) + \delta_{o,t} + \delta_{s,t} + \epsilon_{os,t}, \tag{28}$$

where z_t standardizes each predictor within year over the common estimation sample. Province-year fixed effects account for differences in the overall propensity to designate industries, while sector-year fixed effects account for common industry-year patterns. The comparison therefore relates differences in provincial designations within the same industry and year to differences in revealed priorities, after accounting for province-year characteristics.

Table 4 shows that industries with higher revealed priorities are more likely to be named for support in provincial plans. A one-standard-deviation higher priority is associated with a 3.58 percentage point higher probability of being named when priorities enter alone and 3.57 points when subsidies are included. A one-standard-deviation higher subsidy rate is associated with a 1.13 percentage point higher probability in the joint specification, but the coefficient is not statistically significant. The priority coefficient changes little when subsidies are included, indicating that revealed priorities contain information about stated provincial objectives beyond observed subsidy rates. This is consistent with the distinction between priorities and subsidy rates in the model: priorities are part of the governments' objective function, while subsidies are the general equilibrium outcomes.

Table 4: Provincial Five-Year Plans, priorities, and subsidies

Provincial FYP support (percentage points)	(1)	(2)	(3)
Revealed priority (1 SD)	—	3.58**	3.57**
		(1.35)	(1.32)
Observed subsidy rate (1 SD)	1.17	—	1.13
	(0.77)		(0.76)
Province-year fixed effects	Yes	Yes	Yes
Sector-year fixed effects	Yes	Yes	Yes
Observations	3,847	3,847	3,847

Notes: The sample pools the 43 policy sectors in 2007, 2012, and 2017. Standard errors clustered by province and sector are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Because neither national nor provincial plans enter the recovery, their relationship with priorities provides separate evidence that revealed priorities reflect national and local development objectives. These results are consistent with the incentives of provincial officials. Higher authorities may consider whether industries in national plans have expanded when evaluating officials for promotion, giving provinces an incentive to develop those industries. At the same time, local economic conditions can lead governments to value different industries. Fang, Li, and Lu (2026) find that local industrial support reflects both local economic conditions and upper-level policy priorities. Wang and Yang (2025) show that local officials put more effort into policy experiments initiated by the central government and that the central government's interpretation of the results can affect policies adopted nationwide.

Priorities and industrial growth. Officials who are evaluated on local economic growth may place greater weight on industries that have already grown quickly or that they expect to grow in the future. I relate revealed priorities to manufacturing growth before and after each benchmark. The preceding-growth windows summarize recent local industrial performance, while the subsequent-growth windows describe how industries develop after each benchmark. In both regressions, I relate annualized log real value-added growth to standardized priorities, controlling for the log initial manufacturing share and province and sector fixed effects. The subsequent-growth regression also controls for the observed subsidy rate and prior growth.

Priorities are higher in sectors with faster preceding growth (Table 5, Panel A). A one-standard-deviation higher priority is associated with 4.26 percentage points faster annual growth over 2002–05, 2.36 points over 2007–10, and 2.25 points over 2012–15. All three estimates are statistically significant. In each benchmark year, governments place greater weight on industries that have already grown faster locally. This is consistent with growth incentives of Chinese officials documented by Li and Zhou (2005).

The relationship with subsequent growth is strongest for the 2007 benchmark. A one-standard-deviation higher 2007 priority is associated with 2.82 percentage points faster annual growth over 2008–11, after accounting for subsidies, initial shares, and prior growth. The later coefficients are positive but smaller and statistically insignificant. None of the subsidy coefficients is statistically significant in these specifications. The 2007 result is consistent with governments placing greater weight on industries with stronger growth prospects.

Together, these associations connect revealed priorities to national development goals and local economic performance. In the next section, I perform counterfactual analysis to assess their effects on production and national welfare.

Table 5: Revealed priorities and manufacturing growth

Benchmark year	2007	2012	2017
Panel A: Prior annualized log real-VA growth (p.p.)			
Growth window	2002–05	2007–10	2012–15
Revealed priority (1 SD)	4.26**	2.36***	2.25**
	(1.66)	(0.75)	(0.87)
Observations	866	861	853
Panel B: Subsequent annualized log real-VA growth (p.p.)			
Growth window	2008–11	2013–16	2018–21
Revealed priority (1 SD)	2.82**	0.83	0.28
	(1.24)	(0.62)	(0.38)
Observed subsidy rate (p.p.)	0.027	0.34	0.46
	(1.97)	(0.65)	(0.34)
Observations	850	854	844
Province and sector fixed effects	Yes	Yes	Yes
Log initial VA share	Yes	Yes	Yes
Prior growth in Panel B	Yes	Yes	Yes

Notes: Standard errors clustered by province are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5 Priorities, Production, and Welfare

This section examines how provincial industrial priorities affect production and welfare. I first remove observed subsidies to assess their contribution to industrial similarity and national welfare. I then remove sectoral priorities and allow provincial governments to choose new subsidies to maximize resident welfare without sectoral weights. While both counterfactuals show lower national welfare, the loss is smaller under the welfare-only Nash equilibrium. Removing priorities also shifts subsidies away from manufacturing, reducing activity in industries that generate national productivity spillovers. This shows that revealed priorities partly offset provincial governments' limited incentives to support production whose benefits extend beyond their borders. I then examine the mechanisms behind these effects and their relationship to industrial similarity.

I compute the counterfactual equilibria by expressing the equilibrium conditions in exact changes relative to each calibrated benchmark, following Dekle, Eaton, and Kortum (2008) and Caliendo and Parro (2015). A hat denotes the ratio of a counterfactual value to its value in the benchmark economy. This formulation uses ratios of equilibrium objects to solve for the changes in the counterfactual without the need to separately identify fundamental productivity and bilateral trade costs. Appendix B.4 presents the equations.

To account for national welfare, I define it to be a weighted average of provincial welfare,

where in exact-hat algebra it reads

$$\widehat{\mathcal{W}} = \exp\left(\sum_o \omega_o \log \widehat{W}_o\right), \tag{29}$$

where $\omega_o = \frac{\sum_s V_{os}^b}{\sum_{o,s} V_{os}^b}$ gives the observed GDP weight for each province. Here superscript b denotes the benchmark economy we observe, and reported welfare changes equal $100(\widehat{\mathcal{W}} - 1)$. I also report aggregate real value added, the sum of provincial factor incomes deflated by provincial consumption prices. This measure records real factor earnings before subsidy financing, whereas resident welfare also accounts for subsidy payments and trade imbalances. Aggregate real value added is the unweighted sum across provinces and sectors, whereas the welfare index uses fixed provincial weights to aggregate proportional changes. Appendix D provides the aggregation formulas.

5.1 Removing subsidies

The first experiment sets all subsidy rates to zero and solves for the resulting competitive equilibrium. It compares equilibria under the observed and zero-subsidy schedules without solving a new government policy problem. The experiment asks what the economy would look like without the measured subsidies, holding fundamentals and other unobserved features of the economy fixed.[16] Removing subsidies changes firms' production and input decisions, with effects propagating across provinces and sectors through trade and input-output linkages.

Table 6: Removing all subsidies

	2007	2012	2017
National welfare (percent)	−0.390	−0.225	−0.250
Aggregate real value added (percent)	−2.926	−2.352	−2.633
Largest provincial welfare loss (percent)	−0.990	−0.866	−1.024
Largest provincial welfare gain (percent)	+0.126	+0.305	+0.712
Manufacturing similarity (percent change)	−0.57	−0.22	−0.70
Provinces gaining in welfare	4/30	3/30	5/30

Removing subsidies lowers national welfare by 0.39 percent in 2007, 0.23 percent in 2012, and 0.25 percent in 2017 (Table 6). Aggregate real value added falls by 2.93, 2.35, and 2.63 percent, respectively. The decline in welfare is also widespread across provinces. Only four

[16]This does not imply, however, that this is a counterfactual without any government intervention. As discussed above, governments can influence the economy through other channels, such as land and credit policies. These policies can continue to affect the economy in the absence of subsidies. Removing them would require a model of and data on those policies.

provinces gain from removing subsidies in 2007, three in 2012, and five in 2017. Thus, the observed subsidy schedule benefits residents across much of the country, even though each province finances its own support.

National scale effects determine the welfare consequences of subsidy removal. Holding productivity fixed, removing subsidies raises national welfare by 0.33 percent in 2012, compared with a 0.23 percent decline when national scale effects are active (Appendix G.4). Once productivity is allowed to respond, removing subsidies reduces manufacturing input use and affects the overall efficiency of the economy. There are two main channels through which this occurs. First, contraction lowers the national productivity multiplier Ψ_s in the industries with positive scale elasticities, including for producers in provinces that do not provide the subsidies. Second, lower productivity in upstream industries also raises the cost of intermediate inputs for producers in other industries and provinces. Recall that the calibrated scale elasticities are positive in 25 manufacturing industries. Other industries are affected through changes in intermediate demand, input prices, and factor markets.

Removing subsidies also reduces average inland–coastal manufacturing similarity by 0.57 percent in 2007, 0.22 percent in 2012, and 0.70 percent in 2017. The observed subsidy schedule therefore supports both greater manufacturing scale and a more similar industrial composition across inland and coastal provinces. For comparison, the declines in 2007 and 2012 are each approximately 30 percent of the average annual increase in similarity over the subsequent five years.

5.2 Removing sectoral priorities

The second experiment removes sectoral priorities while allowing provincial governments to continue using subsidies. It asks how governments would choose their policies if they cared only about the welfare of their residents, rather than also valuing production in particular industries. I set $\vartheta_{os} = 0$ for all provinces o and industries s, so provincial governments choose nonnegative subsidies to maximize resident welfare, taking other provinces' policies as given.[17] The resulting policy profile satisfies the local Nash conditions described in Appendix C. In what follows, I compare this counterfactual with the observed 2012 economy and refer to it as the welfare-only counterfactual.

In the welfare-only counterfactual, national welfare falls by 0.07 percent, even though aggregate real value added rises by 0.26 percent. The two measures differ in what they capture and how they are aggregated. In particular, national welfare reflects provincial welfare changes weighted by baseline GDP. As the following figures show, the effects of the

[17]The nonnegative policy domain is a maintained restriction of the reform experiments. Policies in the four background public-service industries remain at their observed values.

counterfactual differ across provinces and sectors.

When governments care only about their residents' welfare, they continue to subsidize production and increase their total spending on subsidies. However, they allocate much less of this support to manufacturing. Manufacturing receives 46.53 percent of observed subsidy expenditure, compared with only 15.39 percent under the welfare-only counterfactual. Support also shifts away from the five industries with highest average priority, whose share of subsidy expenditure falls from 19.91 to 0.19 percent. At the same time, the coastal share rises from 54.53 to 65.61 percent (Table 7). Overall subsidy spending increases from 2.10 to 2.46 percent of observed GDP.

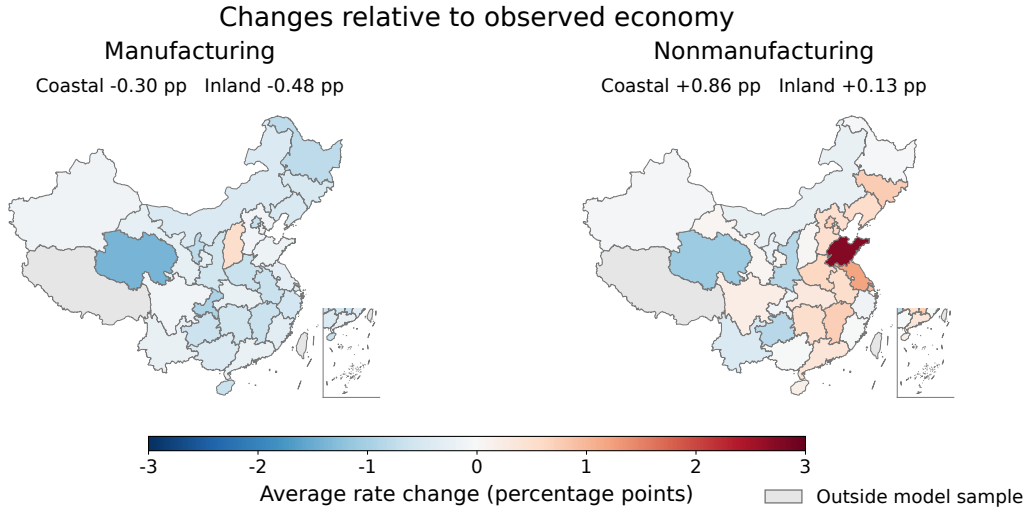
Table 7: Removing priorities reallocates subsidy expenditure, 2012

	Observed	Welfare-only Nash
Manufacturing share	46.53	15.39
Top priority industries' share	19.91	0.19
Coastal share	54.53	65.61
Total expenditure / baseline factor VA	2.10	2.46

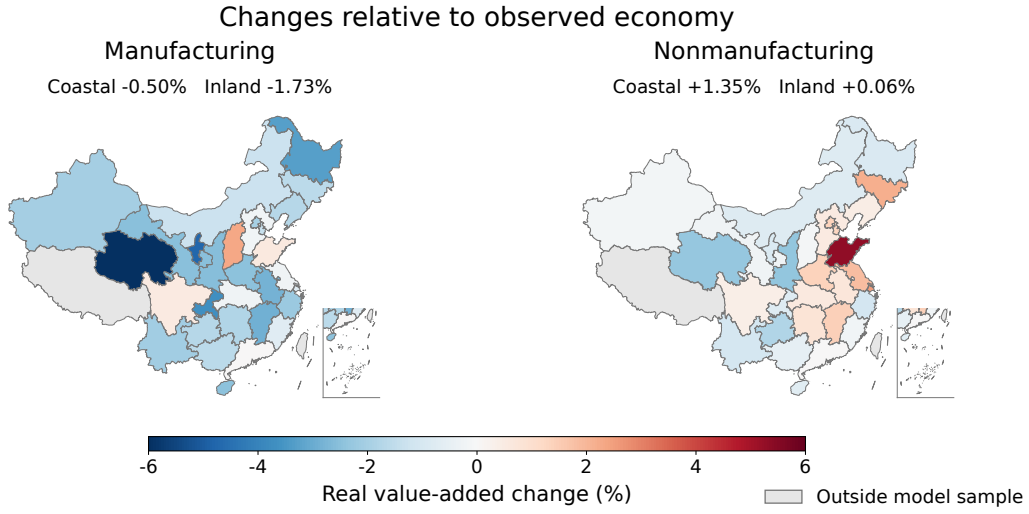
Notes: The five selected priority industries are communication equipment, transport equipment, pharmaceuticals, special machinery, and general machinery.

Figure 5 shows how subsidy rates and real value added change across regions and industries. Manufacturing subsidy rates fall more inland, while the increase in nonmanufacturing subsidy rates is concentrated on the coast. As subsidy rates change, workers and fixed factors reallocate across industries within each province. Manufacturing real value added falls by 0.50 percent on the coast and 1.73 percent inland. In contrast, nonmanufacturing real value added rises by 1.35 percent on the coast and remains approximately unchanged inland, increasing by 0.06 percent. Nationally, manufacturing contracts by 0.91 percent and nonmanufacturing expands by 0.81 percent. The new subsidy schedule shifts economic activity away from manufacturing, where many industries generate national productivity spillovers.

Despite the decline in manufacturing activity, inland–coastal manufacturing similarity rises. This result differs from the first experiment, where removing all subsidies reduces similarity. Here, governments can choose new subsidies to maximize the welfare of their residents. Because the quantitative model uses the same Cobb–Douglas consumption shares across provinces, governments have an incentive to support industries that contribute to local consumption, even though they also trade with other provinces and abroad. The resulting policies reduce overall manufacturing activity while moving provincial production toward a common consumption structure.



(a) Changes in subsidy rates, percentage points



(b) Changes in real value added, percent

Figure 5: Removing priorities reallocates support and production, 2012

Notes: Changes are relative to the observed economy. Nonmanufacturing subsidy rates cover 14 policy sectors; nonmanufacturing value added covers all 18 sectors. Gray areas are outside the sample.

5.3 Provincial welfare and national spillovers

Figure 6 compares the provincial welfare effects of the two counterfactuals in 2012 relative to the observed economy. In both of them, welfare losses are larger inland than on the coast. In the welfare-only counterfactual, inland welfare falls by 0.13 percent and coastal welfare by 0.03 percent, compared with declines of 0.34 and 0.15 percent when all subsidies are removed. Eight provinces gain in the welfare-only counterfactual, compared with three under subsidy removal. Provincial welfare changes in the welfare-only counterfactual range from a loss of 0.75 percent to a gain of 0.51 percent. Allowing governments to maximize resident welfare by choosing new subsidies reduces the aggregate and regional welfare losses, though inland provinces continue to experience larger losses.

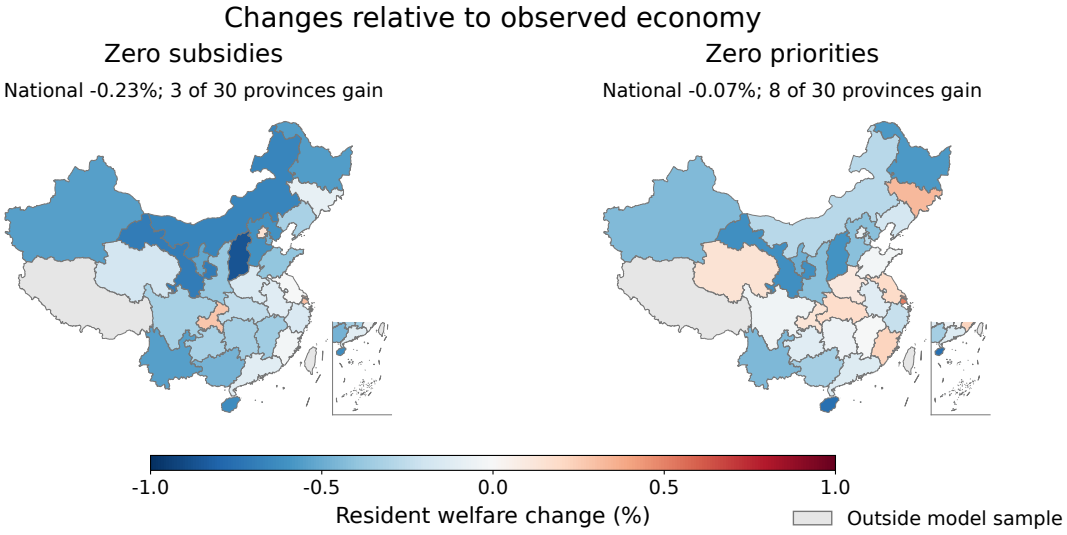


Figure 6: Provincial welfare effects of removing subsidies and priorities, 2012
Notes: Percent changes in resident welfare relative to the observed economy. The left panel removes subsidies; the right panel removes sectoral priorities.

The difference between provincial and national incentives helps explain why the revealed priorities increase national welfare relative to the welfare-only counterfactual. When a provincial government supports an industry with a positive scale elasticity, the expansion lowers the price of its output and generates productivity spillovers. Other industries and provinces benefit from cheaper intermediate inputs. At the same time, producers in the supported industry benefit from higher national productivity, where the size of the multiplier depends on the industry's scale elasticity. Provincial governments value these effects only to the extent that they benefit their own residents. The revealed priorities give governments an additional incentive to support particular industries and, in this counterfactual, partly offset these limited incentives by sustaining manufacturing activity.

5.4 Welfare decomposition and sensitivity

To account for the sources of welfare changes, I integrate the terms-of-trade, scale-externality, and subsidy-distortion components in (20) along the change in subsidies. The contributions are expressed in percentage points and sum to the national welfare change in (29). Appendix G.7 presents the derivation and integration procedure.

To understand the terms-of-trade effect here, note that governments value changes in the prices at which their provinces sell output and purchase goods, alongside productivity gains and the resource costs created by subsidies. These price incentives have been discussed heavily in trade-policy analysis (Ossa 2014, Lashkaripour and Lugovskyy 2023). Positive sectoral priorities add a motive to subsidize local activity, including expansion whose price effects are unfavorable for resident welfare. Removing priorities eliminates this additional motive and changes the allocation of subsidies. The resulting terms-of-trade contribution measures the price consequences of that adjustment.

Table 8 applies the welfare decomposition to the two counterfactuals in 2012. Compared with removing subsidies, removing priorities gives a larger terms-of-trade effect. Under welfare-only competition, local governments use subsidies to manipulate trading prices to improve resident welfare. The loss from scale externalities is smaller when priorities are removed because governments continue to subsidize some industries with positive scale elasticities, although total manufacturing subsidy is lower than under the revealed priorities. Removing all subsidies improves the subsidy-distortion contribution by eliminating the wedges. Removing priorities does not: the reallocation of physical inputs under the remaining wedges contributes negatively to welfare. The revealed priorities sustain productive scale and a more favorable allocation of inputs under subsidy wedges. Together, these benefits outweigh their unfavorable terms-of-trade effect.

Table 8: Welfare effects of removing subsidies and priorities, 2012

	Terms of trade	Scale externalities	Subsidy distortions	Total
Remove subsidies	+0.507	−0.771	+0.039	−0.225
Remove priorities	+0.730	−0.709	−0.089	−0.068

Notes: Contributions are in percentage points relative to the observed economy and sum to the national welfare change.

The welfare comparison is sensitive to the calibration. Assigning a scale elasticity of 0.17 to industries with zero baseline elasticities broadens the productivity benefits of industrial support. Expansion in these industries now raises productivity, benefiting residents through consumption and producers through intermediate inputs. Provincial governments value these

benefits even when they place no additional weight on sectoral activity. The alternative calibration therefore changes the relative returns to the activities supported in the welfare-only equilibrium. In 2012, both manufacturing and nonmanufacturing real value added increase when priorities are removed, and national welfare rises by 0.462 percent. Removing all subsidies instead lowers welfare by 0.716 percent, because it also removes support for activities generating these productivity gains. The comparison shows that the welfare value of the observed priorities depends on which industries generate external benefits, as well as on their strength (Appendix Table G.7).

The geographic reach of externalities changes the relationship between provincial and national incentives. Under national externalities, expanding an industry's input use in one province raises productivity in that industry throughout China, while the provincial government values the resulting benefits only if they affect its own residents. Provincial externalities bring the direct technological benefits within the producing province, reducing the cross-province productivity benefit that additional sectoral priorities can help generate. In this calibration, removing subsidies and removing priorities raise national welfare by 0.300 and 0.335 percent in 2012, respectively. Appendix Table G.6 reports the remaining sensitivity exercises, including annual production and expenditure shares and alternative foreign-trade elasticities.

With the above discussion in mind, I retain national scale elasticities as the preferred specification, supported by the estimated manufacturing elasticities and the evidence summarized in Section 3.

The observed allocation delivers higher national welfare than the allocations in either baseline counterfactual. These baseline comparisons show that the revealed priorities sustain production with benefits beyond individual provinces. The next section asks whether reallocating provincial priorities can raise national welfare further within existing fiscal resources.

6 Designing Provincial Priorities

The previous section shows that revealed provincial priorities capture governments' incentives to support industries beyond the benefits they provide to local residents. This raises a further question: could a different set of priorities raise national welfare? To answer this question, I retain the policy instruments and local financing arrangements in the model and redesign the priorities. The exercise proceeds in two steps. First, I solve the national planner's problem, maximizing national welfare subject to each province's observed real fiscal constraint. Second, I construct the priorities that support the resulting subsidy schedule as a

Nash equilibrium. The redesigned priorities therefore provide incentives for local governments to choose the constrained-efficient subsidies under Nash competition. In what follows, I refer to the priorities inferred from observed policies as "revealed priorities" and those constructed to implement the constrained-efficient allocation as "redesigned priorities."

6.1 A constrained national policy benchmark

Following the welfare criterion in the previous section, I define the national planner's objective as

$$G(\boldsymbol{\tau}) = \sum_{o} \omega_o \log W_o(\boldsymbol{\tau}). \tag{30}$$

The planner chooses provincial subsidy rates subject to competitive equilibrium, nonnegative subsidy rates, and a separate real-spending constraint for each province:

$$\max_{\boldsymbol{\tau}} G(\boldsymbol{\tau}) \quad \text{subject to} \quad F_o(\boldsymbol{\tau}) \equiv \frac{\sum_s \tau_{os} B_{os}(\boldsymbol{\tau})}{P_o^c(\boldsymbol{\tau})} \leq F_o(\boldsymbol{\tau}^b), \qquad o \in \mathcal{R}. \tag{31}$$

The constraint ensures that each province's real subsidy spending does not exceed its observed level. The numerator is total provincial subsidy expenditure, evaluated at the production levels induced by the policy schedule. Deflating by the provincial consumption price index gives the real value of this expenditure. The planner reallocates subsidies across industries within each province's observed real subsidy-spending ceiling. The resulting subsidy schedule $\boldsymbol{\tau}^*$ implements the constrained-efficient allocation of the observed fiscal resources. I solve the planner's problem numerically to obtain the constrained-efficient subsidy schedule.[18]

Table 9: Priority redesign within observed provincial real budgets, 2012

	Change from observed economy
National resident welfare	+1.276
Aggregate real value added	+1.091
Real spending in every province	0.000
Smallest provincial welfare change	−0.055
Largest provincial welfare change	+2.003
Provinces gaining in welfare	28/30

Notes: Changes are in percent, except the number of provinces gaining.

The constrained-efficient equilibrium raises national welfare by 1.28 percent and aggregate real value added by 1.09 percent (Table 9). All provincial fiscal constraints bind, so each

[18]The computed schedule satisfies the planner's first-order and constrained second-order sufficient conditions. Optimization from zero, uniform, and randomized subsidies under tighter stopping criteria yields essentially identical welfare gains and closely agreeing policies. Appendix E describes the numerical comparisons.

province's real subsidy spending equals its observed level. The welfare improvement comes from changing how existing resources are allocated across industries, rather than from increasing real subsidy spending. Welfare increases in 28 of the 30 provinces, with provincial changes ranging from -0.06 to $+2.00$ percent. Yunnan and Qinghai experience small welfare losses. The gains extend beyond the provinces and industries receiving additional support. Changes in production affect national productivity, prices, and demand for intermediate inputs, transmitting the effects of the reallocation across the economy.

6.2 Redesigning provincial priorities

The constrained-efficient subsidy schedule improves national welfare, but it can be implemented through decentralized policy only if provincial governments have incentives to choose it under the Nash equilibrium. Recall the procedure developed in Section 4 to recover the governments' priorities. I apply the procedure here, by replacing the observed subsidy schedule with the constrained-efficient schedule and solving each government's first-order conditions for the redesigned priorities $\boldsymbol{\vartheta}_o^*$,

$$a_o(\boldsymbol{\tau}^*) + \mathcal{B}_o(\boldsymbol{\tau}^*)\boldsymbol{\vartheta}_o^* = 0. \tag{32}$$

As in the recovery exercise, a_o collects the marginal welfare benefit from adjusting subsidies, while $\mathcal{B}_o$ collects the effects on local sectoral real value added. The redesigned priorities make the government's additional valuation of changes in local industry offset the resident-welfare incentive to move away from the constrained-efficient subsidy schedule. The inversion thus translates the coordinated allocation into the sectoral objectives that induce each province to choose its part of the policy schedule.

In the two step process, the fiscal constraints facing the national planner and the provincial governments are different. The national planner chooses a subsidy schedule subject to each province's observed real spending constraint. Hence the welfare gain is from the reallocation of existing resources, not from additional spending. However, provincial deviations are evaluated under the original local financing rule, allowing both the allocation of subsidies and total spending to adjust. The redesigned priorities give each local government an incentive to choose its part of the constrained-efficient schedule.

Following Ossa (2014) and Ferrari and Ossa (2023), I assess these priorities using the provincial optimality conditions and numerical checks. The first-order and second-order conditions for each province's full subsidy vector establish strict local optimality for all 30 governments. Checks of feasible finite deviations find no profitable deviation. Sequential provincial policy adjustment from the observed profile and a perturbed subsidy schedule

also returns to a neighborhood of $\boldsymbol{\tau}^*$. Together, these checks provide numerical support for implementation as a Nash equilibrium.

To understand the role of the redesigned priorities, I compare the national planner's and provincial governments' incentives to subsidize an industry. Consider an interior subsidy rate τ_{ok}, chosen by province o for industry k. The provincial government's first-order condition is

$$\frac{\partial \log W_o}{\partial \tau_{ok}} + \sum_s \vartheta^*_{os} \frac{\partial \log V_{os}}{\partial \tau_{ok}} = 0,$$

where the first term captures the effect of the subsidy on the welfare of local residents and the second captures the effects on local industries. The national planner's first-order condition for the same subsidy is

$$\sum_r \omega_r \frac{\partial \log W_r}{\partial \tau_{ok}} - \sum_r \lambda_r \frac{\partial F_r}{\partial \tau_{ok}} = 0,$$

where F_r is province r's real subsidy spending and λ_r is the shadow value of its real fiscal constraint. The planner accounts for the welfare effects across all provinces and for changes in their real subsidy spending.

Combining the two conditions gives

$$\omega_o \sum_s (\mathcal{B}_o)_{ks} \vartheta^*_{os} = \sum_{r \neq o} \omega_r \frac{\partial \log W_r}{\partial \tau_{ok}} - \sum_r \lambda_r \frac{\partial F_r}{\partial \tau_{ok}}. \tag{33}$$

Both the provincial government and the planner account for the effects on province o's residents. The first term on the right-hand side therefore captures the welfare effects on residents of other provinces, while the second accounts for changes in real subsidy spending across all provinces. The left-hand side shows how the redesigned priorities provide the provincial government with the additional incentive to choose the constrained-efficient subsidy.

Intuitively, the redesigned priorities give provincial governments incentives to account for the national consequences of their policies. Suppose a provincial government places a priority on only one local industry. When choosing its subsidy schedule, the government considers how each subsidy affects that industry, in addition to its effects on local residents' welfare. If the government has priorities for several industries, it considers how each subsidy affects all of them. Subsidies also affect welfare in other provinces through trade, input-output linkages, and relative prices. The local government accounts for these general equilibrium effects, but values them only to the extent that they affect its own residents. The redesigned priorities are chosen jointly so that, for each subsidy, the government's additional valuation of changes across local industries reflects the welfare effects on other provinces, adjusted for the effects on provincial fiscal constraints.

Equation (20) shows how the redesigned priorities improve national welfare. The terms-of-trade and scale-externality channels contribute 0.750 and 0.727 percentage point, respectively, while the subsidy-distortion channel contributes -0.201 percentage point. The redesign accepts a larger distortion cost from the allocation of inputs under subsidy wedges because the resulting productivity and price benefits are higher. Scale externalities and subsidy distortions together contribute a net production-efficiency gain of 0.526 percentage point, producing an overall welfare gain of 1.276 percent. These gains arise from reallocating support within unchanged provincial real budgets. The negative distortion contribution therefore reflects the resource-allocation effects of the redesigned subsidies, rather than an increase in real spending. Appendix Table G.10 reports the decomposition for all three years.

The welfare gain from redesign persists across the alternative calibrations in Appendix G.6. In 2012, the gain ranges from 0.849 percent under provincial scale externalities to 2.004 percent when positive scale elasticities are assigned to industries with zero baseline elasticities (Appendix Table G.7).

6.3 How priorities and production change

I next compare the redesigned priorities with the revealed priorities and examine the resulting changes in production. Figure 7 shows the four industries with the largest increases and the four with the largest decreases in average priority. Each point represents a province, and the right-hand column reports the change in national real value added for each industry under the full set of redesigned priorities. The redesign raises priorities for several industries that supply basic manufactured inputs, while reducing priorities for a range of manufacturing and nonmanufacturing industries. Within each industry, priority changes often have the same direction across provinces, but their magnitudes differ. The redesigned priorities therefore give governments a common direction for reallocating support, while allowing the strength of the incentive to vary across provinces.

Chemicals, nonferrous metals, paper, and ferrous metals have the largest increases in average priority. Priorities for chemicals and nonferrous metals rise in all 30 provinces, while the priority for ferrous metals rises in 20 provinces. The priority for paper rises in 28 of the 29 provinces where the industry is active. National real value added in these industries increases by 12.34, 24.69, 15.87, and 5.43 percent, respectively.[19] These industries supply manufactured inputs used by other industries, so their expansion can affect production costs and activity throughout the economy. Under the redesigned priorities, subsidy expenditure

[19]The model does not impose industry-specific natural resource or capacity constraints, which may limit the expansion of some upstream industries. In the presence of such constraints, the actual increases in value added would be smaller.

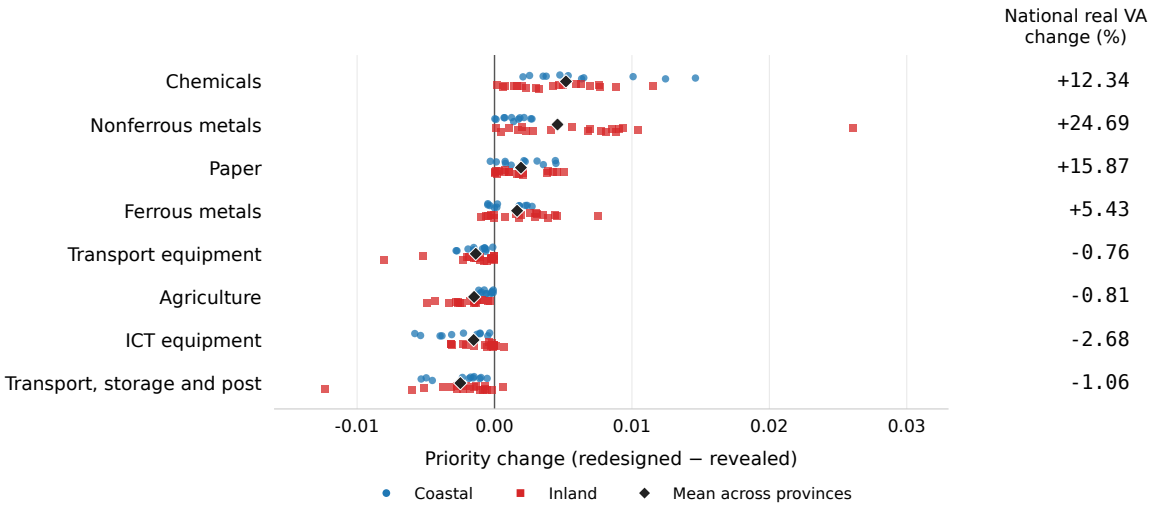


Figure 7: Revealed and redesigned provincial priorities, 2012

Notes: Points represent provinces; diamonds show sector means. The industries shown have the largest positive and negative mean priority changes. Value-added changes are under the full reform.

is directed toward these industries. In contrast, average priorities decline in agriculture, transport services, transport equipment, and ICT, among other industries.

The redesigned priorities remain closely related to the revealed priorities. Across the 1,285 active province–industry pairs, the correlation between their levels is 0.978. At the same time, 972 priorities decline and 313 increase. The changes reported above are relative to each province's revealed priorities. An increase in the priority for chemicals, for example, does not mean that chemicals becomes the highest-priority industry in every province. The redesign redirects sectoral incentives while retaining much of the revealed priority structure.

Relative to revealed priorities, the redesign shifts manufacturing priorities toward upstream industries. These industries supply inputs that are used in other industries, so changes in their production costs can affect activity throughout the economy. To examine this relationship, Figure 8 compares changes in priorities and national real value added with a measure of industry upstreamness. Following Antràs, Chor, Fally, and Hillberry (2012), upstreamness is the expected number of production stages before an industry's output reaches final use. I calculate this measure using domestic input-output sales across all 47 industries, with exports treated as final uses. The figure shows the 29 manufacturing industries, with each industry given equal weight. For each industry, the change in priority is averaged across the provinces where it is active.

Both priority adjustments and real-value-added growth are positively associated with upstreamness. The correlation between mean priority changes and upstreamness is 0.61 and remains positive when chemicals and nonferrous metals are omitted, showing that the

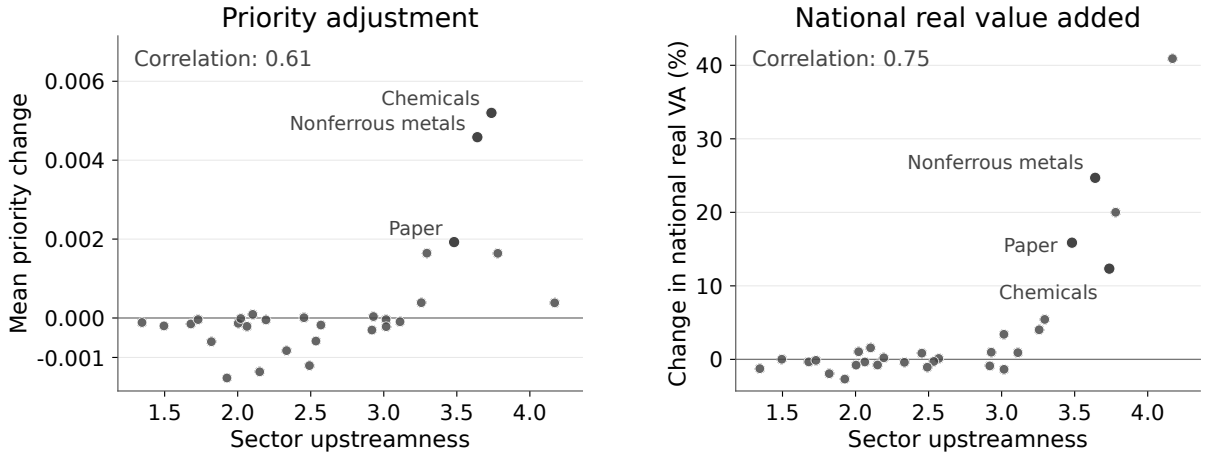


Figure 8: Priority redesign and the position of manufacturing industries in production
Notes: Each point represents a manufacturing industry. Changes compare the constrained-efficient equilibrium with the observed 2012 economy.

association extends beyond the two most prominent increases. The reform increases the importance of industries that supply inputs through longer domestic production chains. Their expansion changes intermediate demand and transmits productivity gains through domestic supply relationships. Upstreamness describes this common feature of the jointly chosen sectoral adjustments. The increase in upstream priorities persists across alternative calibrations: the correlation between upstreamness and average priority changes ranges from 0.45 to 0.62 (Appendix Table G.9).

The welfare gains coexist with little change in the average similarity of provincial industrial composition. In the constrained-efficient equilibrium, the inland–coastal manufacturing cosine changes slightly, from 0.5780 to 0.5786, even though sectoral activity changes substantially. Several expanding upstream industries increase their composite inputs in most provinces with active production, and the resulting growth in their national input raises productivity. The Nash equilibrium under redesigned priorities therefore changes production levels and national industry scale while leaving the average similarity of provincial industry mixes almost unchanged. Its welfare gains come from which industries expand, the productivity benefits they generate, and the resources used to support them.

7 Conclusion

This paper provides a framework to recover provincial governments' industrial priorities from observed subsidies and production. Subsidy rates alone do not measure these priorities

because they also reflect differences in economic fundamentals and the returns to industrial support. A spatial political economy model accounting for trade, production networks and national external economies of scale allows me to recover the additional value governments attach to local sectoral development through a revealed-preference approach.

I find that the revealed priorities become more aligned between inland and coastal provinces, indicating that their governments have become more similar in the industries they want to develop. Higher priorities in industries identified for development in national Five-Year Plans connect provincial objectives to a common development agenda. Comparing these priorities with provincial Five-Year Plans, I find that a one-standard-deviation higher priority is associated with a 3.57 percentage point higher probability that an industry is designated for support, even after controlling for observed subsidy rates. Because neither national nor provincial plans enter the recovery, these relationships support interpreting the revealed priorities as government development objectives. Priorities are also higher in industries with stronger preceding growth, consistent with governments placing greater weight on activities that have expanded faster locally in the past few years.

In the model, provinces finance subsidies locally, while national external economies of scale, trade and production networks allow local industrial expansion to benefit residents in other provinces. I find that the revealed priorities partly offset provincial governments' limited incentives to support industries whose benefits extend beyond their borders. Giving additional weight to local industrial development helps sustain manufacturing activity that receives less support when governments maximize their residents' welfare alone. These results show that increasing similarity in industrial composition need not imply wasteful duplication. Assessing provincial industrial policies requires accounting for the national benefits and costs of the activities they support.

The composition of provincial priorities can nevertheless be improved. I show that redesigned priorities implement a constrained-efficient allocation that raises national welfare by 1.28 percent in 2012, benefits 28 of the 30 provinces, and keeps each province's real subsidy spending at its observed level. Support shifts toward several upstream manufacturing industries, whose productivity gains reach other industries through intermediate inputs. This reallocation improves welfare with little change in average industrial similarity. The exercise shows how changing provincial objectives can give local governments incentives to choose policies with greater national benefits while retaining local policy choice and financing.

The analysis leaves several questions for future research. The model is static, so the recovery does not distinguish governments' valuation of current industrial activity from their expectations about its future benefits. A dynamic extension with investment, learning, and adjustment costs could examine how provincial policies shape comparative advantage over

time. Section 4.5 relates the revealed priorities to development plans and local industrial growth, providing evidence consistent with political incentives. The model does not, however, explain how these incentives shape provincial priorities. Future work could study how central authorities influence provincial objectives through development plans, personnel decisions, and fiscal transfers, and how these influences interact with local economic conditions. This could help explain how redesigned priorities might be implemented in practice.

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A Data Construction and Empirical Specifications

A.1 Province and sector coverage

The production panel and quantitative model cover 30 provincial-level regions, excluding Tibet. The coastal group consists of Beijing, Tianjin, Hebei, Liaoning, Shanghai, Jiangsu, Zhejiang, Fujian, Shandong, Guangdong, and Hainan. The remaining nineteen provinces form the inland group. There are 55 unordered coastal–coastal pairs, 171 inland–inland pairs, and 209 inland–coastal pairs, for a total of 435 distinct pairs. Regional and pairwise averages give equal weight to each province or pair, respectively.

Annual industrial data are harmonized to a broad classification based on the 2002 industries. Removing mining yields the 29-sector manufacturing sample used in the descriptive production and growth comparisons. The quantitative classification contains 47 sectors. Four public-service sectors are excluded from provincial sectoral objectives: water, environment and public facilities, education, health and social work, and public administration and social security. All 47 sectors enter the economy's production, expenditure, and market-clearing accounts. The revealed-priority and provincial-plan comparisons cover the remaining 43 policy sectors, including manufacturing and nonmanufacturing activities. The active set contains industries with positive baseline production and is held fixed within each experiment. There are 1,288, 1,285, and 1,286 active policy cells in 2007, 2012, and 2017, respectively.

A.2 Constructed real value added

The annual production series uses operating revenue from the China Industry Statistical Yearbook, with industrial sales used when operating revenue is unavailable. Publication year $t+1$ is assigned to economic year t. After harmonizing industry names and classification changes, constructed real value added is

$$\widetilde{VA}_{ost} = \frac{\text{Revenue}_{ost}}{PPI_{st}}\,\nu_s^{2007}, \qquad \nu_s^{2007} = \left(\frac{VA_s}{Y_s}\right)_{\text{national IO,2007}}. \tag{A.1}$$

The national sectoral producer-price index is expressed as a price ratio. The first economic year, 2000, uses the 2001 index, the first year of the chained series. When several detailed input-output industries map to one empirical sector, ν_s^{2007} is their summed value added divided by summed gross output. The common coefficient converts sectoral revenue to value added while preserving relative revenue across provinces within the same industry. It holds the sectoral value-added/output ratio fixed across provinces and dates.

Some industries appear separately only in later yearbooks. The harmonization retains

reported detailed revenues and allocates the difference between reported provincial total revenue and the sum of reported detailed industries across the unreported categories. Allocation weights are proportional to predicted revenue from a pooled regression of log revenue on log(Employment + 1) and industry fixed effects, estimated where the relevant industries are observed. Unavailable employment entries take the value zero in this predictor. The published total determines the amount allocated. Fitted revenue determines its distribution. Any remaining discrepancy when all candidate industries are reported remains an unallocated residual. Deflation and the input-output conversion follow this step.

The descriptive periods are 2000–02, 2003–05, 2006–08, 2009–11, 2012–14, and 2015–18. Provincial sectoral shares are computed after averaging real value added within each period. The final period contains 2015, 2016, and 2018. 2017 is not interpolated. The activity series extends beyond 2018 for the subsequent-growth regressions. Empirical constructed real value added uses producer-price deflation, whereas model real value added is factor income deflated by the provincial consumption price index.

A.3 Similarity measures

Let $\widetilde{VA}_{os,p}$ denote constructed real value added in province o and sector s, averaged over period p. The manufacturing-share vector is

$$x_{os,p} = \frac{\widetilde{VA}_{os,p}}{\sum_{j\in\mathcal{M}} \widetilde{VA}_{oj,p}}, \qquad \mathbf{x}_{o,p} = (x_{os,p})_{s\in\mathcal{M}}, \tag{A.2}$$

where $\mathcal{M}$ contains the 29 manufacturing industries. Similarity between two vectors is measured by the cosine in equation (1).

The fixed coastal benchmark is constructed in three steps. First, real value added is averaged over 2006–08 separately for each province and manufacturing sector. Second, these averages are converted to provincial manufacturing shares. Third, the eleven coastal share vectors are averaged with equal province weights. Every subsequent provincial vector is compared with this same benchmark. The regional lines in Figure 3 average the resulting provincial cosines equally. Pairwise production cosines instead compare two provincial vectors in the same period. Both measures use the 29 manufacturing sectors and equation (1).

For priority similarity, the vectors contain signed revealed priorities. The sector set for each pair is the intersection of sectors active in both provinces at all three benchmark dates. Holding this intersection fixed prevents changes in estimated sector coverage from changing the comparison across years. The main figure uses the 43 policy sectors, and Appendix Table F.2 also reports manufacturing. Signed priorities permit negative cosines in principle.

A cosine measures alignment and is invariant to a common positive scaling of either priority vector.

The scale-elasticity comparison in Figure F.4 divides manufacturing at the median calibrated elasticity, 0.14. Fourteen industries have $\phi_s > 0.14$ and fifteen have $\phi_s \leq 0.14$. Both the provincial vector and the fixed coastal benchmark are restricted to the same group before calculating its cosine. The all-manufacturing series is identical to the inland series in Figure 3.

A.4 Fiscal support in the China Tax Survey

The China Tax Survey reports subsidy income, domestic VAT refunds, VAT reductions, business-tax relief, corporate-income-tax relief, and corporate-income-tax credits. The questionnaire fields are harmonized separately by year. Subsidy income includes VAT refunds, while a separate field also records those refunds. Let C_{it} denote reported subsidy income and V_{it} separately reported domestic VAT refunds. For nonnegative reports I use $\max\{C_{it}, V_{it}\}$ to avoid adding overlapping amounts. Other separately reported relief and credit components are then added. Ordinary export VAT refunds and the separate VAT loan-repayment rebate are excluded.

For the quantitative analysis, the signed construction retains reported reversals and adjustments. Its subsidy-income/VAT term is

$$\max\{C_{it}, V_{it}, 0\} + \min\{C_{it}, V_{it}, 0\}. \tag{A.3}$$

The remaining relief components enter with their reported signs, and missing component fields enter as zero. For nonnegative observations, equation (A.3) reduces to the maximum rule. This rule treats the two reports as overlapping to the greatest extent consistent with their recorded amounts. It is a measurement convention for reconciling overlapping reports. Ordinary tax payments are not deducted from support.

One report is retained per normalized taxpayer identifier by selecting the record with the most complete relevant fields, then the largest output, then the earliest source row. Firms must have mapped province and sector codes and positive finite output. When revenue is positive and the output/revenue ratio is finite, that ratio must lie between 0.001 and 1,000. Other records are retained only if output is no more than 10,000 times the sector median among records with positive revenue and a finite output/revenue ratio. The national median supplies the fallback when necessary.

The upper-tail screen excludes firms above the annual-sector ninety-ninth percentile of positive nonnegative-support/output ratios. A national cutoff is used when a sector has fewer

than fifty positive observations. Annual province-sector cells must also have a nonnegative support/output ratio no larger than 20 percent. These screens determine the retained sample. Signed support is then calculated on the same observations, without a separate negative-tail deletion.

For each model policy window, support and output are summed across retained firms and years before constructing their ratio. The model's effective subsidy expresses this payment relative to the gross input bill, as specified in Section 4. The observed signed policy wedges range from -0.003876 to 0.159007. Twelve active policy cells without retained pooled observations receive zero rates in the structural inputs and remain explicitly flagged as imputed. Four occur in 2007: Shanxi tobacco, Inner Mongolia cultural and sporting goods, Gansu tobacco, and Qinghai cultural and sporting goods. Eight occur in 2017: chemical fibers in Beijing and Shanxi, cultural and sporting goods and chemical fibers in Inner Mongolia, tobacco in Jiangxi, measuring instruments in Hainan, and cultural and sporting goods and chemical fibers in Ningxia. These cells are excluded from the observed-rate sample used in the provincial-plan regressions. This preserves the distinction between an observed zero subsidy and an unavailable policy observation.

Figure 2 uses the nonnegative support measure and positive survey revenue. Its right panel divides regional support by revenue times the fixed national 2007 ICT value-added/output coefficient, using the retained province-sector eligibility and original support/output screens. Its left panel uses the separate industry-yearbook measure of the regional ICT share in manufacturing. Both panels aggregate amounts before taking ratios.

A.5 Benchmark timing

Table A.1 records the input windows behind the three benchmark labels. Manufacturing activity is averaged around the input-output benchmark, while nonmanufacturing activity comes from the benchmark accounts. The 2017 policy vector uses the latest available survey window, 2015–16. These assignments also determine the interpretation of the surrounding growth windows.

Table A.1: Input windows for the benchmark economies

Benchmark	Manufacturing activity	Input-output table	Policy window
2007	Mean, 2006–08	2007	Pooled, 2007–08
2012	Mean, 2011–13	2012	Pooled, 2011–13
2017	Mean, 2015, 2016, 2018	2017	Pooled, 2015–16

A.6 National-plan designations

The national-plan indicator maps industries, products, and technologies identified for development in the Eleventh, Twelfth, and Thirteenth Five-Year Plans to the empirical sector classification. The concordance retains the plan category, source document, passage locator, mapping weight, and mapping explanation. Within a sector and benchmark year, the support indicator equals one when the sum of recorded mapping weights is positive. Repeated references to an activity therefore leave its binary support status unchanged. The industry mapping is constructed independently of revealed priorities.

The manufacturing concordance designates 13, 15, and 14 of the 29 sectors in 2007, 2012, and 2017, respectively. The broader concordance designates 27, 28, and 28 of the 43 policy sectors.

In each national regression, priorities are standardized using the population standard deviation across eligible finite priority observations in that year and set of industries. The national indicator remains in its raw zero–one units. Regressions are unweighted, include province fixed effects, and use sector-clustered HC1 standard errors with the finite-cluster adjustment. The reported stars use two-sided normal-reference p-values. Thus each coefficient is the conditional priority gap between supported and unsupported sectors, rather than the response to a one-standard-deviation change in support.

A.7 Provincial-plan designations

The CNRDS Industrial Policy Research Database supplies the province, plan, named activity, quoted passage, historical industry codes, and separate local and national key-support flags. The analysis uses the local key-support flag. Activities are matched to the 43 policy sectors before aggregation. Identical province–plan–passage–sector links are deduplicated, and a sector receives local support when at least one retained link carries an affirmative local key-support flag. Repetition of a passage does not increase its regression weight.

Industry matching uses both the supplied codes and named activities. It distinguishes equipment manufacturing from network services, pharmaceutical production from health services, and material recovery from waste disposal. Corrections to sector assignments preserve the provider's support labels. The mapping contains 5,155 record–sector links and 5,029 distinct passage–sector links drawn from 3,065 source activity records. A zero indicates that no retained matched passage records local key support.

The pooled observed-rate sample contains 3,847 cells: 1,284 in 2007, 1,285 in 2012, and 1,278 in 2017. The twelve zero-imputed policy observations are excluded, while observed zero and negative rates remain. The subsidy-only, priority-only, and joint specifications use

this same sample. Both predictors are standardized within year over these observations. Province-year and sector-year fixed effects are included, and standard errors are clustered by province and sector with $t(29)$ inference.

A.8 Manufacturing growth specifications

The outcome for a growth window $[a, b]$ is $100[\log \widetilde{VA}_{osb} - \log \widetilde{VA}_{osa}]/(b-a)$. The prior-growth windows are 2002–05, 2007–10, and 2012–15. Subsequent-growth windows are 2008–11, 2013–16, and 2018–21. Priorities are standardized within benchmark year over finite manufacturing priorities before outcomes are matched. Eligible manufacturing cells with available outcomes and required controls enter each regression.

Both specifications include province and sector fixed effects and the log of the sector's initial share in provincial manufacturing. The subsequent-growth specification also includes prior growth and the observed subsidy rate in percentage points. Regressions are unweighted, with standard errors clustered by province. The estimation samples contain 866, 861, and 853 observations for prior growth and 850, 854, and 844 for subsequent growth. Table 5 reports the coefficients, standard errors, and significance levels for these windows.

B Competitive Equilibrium and Calibration

This appendix derives the competitive equilibrium of the domestic model in Section 3, then introduces the foreign-trade extension used in the quantitative analysis and its exact-change representation. All production and occupational sums are over the fixed active set. Fundamental productivities, input shares, preferences, endowments, bilateral trade costs, and the coefficients converting provincial inputs into national units are fixed within an experiment.

B.1 Firm optimization and Armington demand

Write $k_s = \alpha_s(1-\gamma_s)$ and $\ell_s = (1-\alpha_s)(1-\gamma_s)$. A producer minimizes $r_o K + w_{os} L + \sum_j P_{oj} M_j$ subject to producing one unit of its Cobb–Douglas input composite. The first-order conditions equate expenditure shares to exponents:

$$r_o K_{os} = k_s B_{os}, \qquad w_{os} L_{os} = \ell_s B_{os}, \qquad P_{oj} M_{os,j} = \gamma_{sj} B_{os}.$$

Substitution into the input requirement gives

$$u_{os} = \left(\frac{r_o}{k_s}\right)^{k_s} \left(\frac{w_{os}}{\ell_s}\right)^{\ell_s} \prod_{j:\gamma_{sj}>0} \left(\frac{P_{oj}}{\gamma_{sj}}\right)^{\gamma_{sj}}. \tag{B.1}$$

The cost-share constant in equation (5) is therefore $C_s = k_s^{-k_s}\ell_s^{-\ell_s}\prod_{j:\gamma_{sj}>0}\gamma_{sj}^{-\gamma_{sj}}$. The subsidy multiplies every privately paid input cost by the same factor $1-\tau_{os}$ and leaves these conditional shares unchanged. Profit maximization at a positive output level gives equation (7). Multiplying by output gives the producer accounts in equation (6).

For delivered origin prices $p_{ro,s} = \kappa_{ro}^{s} p_{os}$, a composite buyer minimizes $\sum_o p_{ro,s}x_{ro,s}$ subject to the CES quantity requirement. The first-order conditions and homogeneity imply

$$x_{ro,s} = Q_{rs}\left(\frac{p_{ro,s}}{P_{rs}}\right)^{-\sigma_s}, \qquad P_{rs} = \left(\sum_o p_{ro,s}^{1-\sigma_s}\right)^{1/(1-\sigma_s)}. \tag{B.2}$$

Thus expenditure shares are $p_{ro,s}^{1-\sigma_s}/\sum_m p_{rm,s}^{1-\sigma_s}$, yielding equation (15) when origin sums run over the provinces in $\mathcal{R}$. This is the standard unweighted Armington composite. Final consumers and intermediate users face the same sectoral price index.

B.2 Factor clearing and household accounts

For a worker in province o, the distribution of maximum earnings is

$$\Pr\left(\max_s w_{os}\epsilon_{hs} \le y\right) = \exp\left(-y^{-\zeta}\sum_s w_{os}^{\zeta}\right).$$

Its mean is y_o in equation (10). Integrating over the event that occupation s offers the highest earnings gives the choice share λ_{os} and effective labor supply in that equation. Mean realized earnings conditional on the chosen occupation equal y_o. Accordingly, a sector's share of provincial labor income equals its employment share.

Let $I_o^L = \sum_s \ell_s B_{os}$ and $I_o^K = \sum_s k_s B_{os}$. Factor clearing implies $y_o = I_o^L/\bar{L}_o$ and $r_o = I_o^K/\bar{K}_o$. With $\Gamma_\zeta = \Gamma(1-1/\zeta)$, efficiency wages are recovered from bills as

$$w_{os} = \frac{y_o}{\Gamma_\zeta}\left(\frac{\ell_s B_{os}}{I_o^L}\right)^{1/\zeta}. \tag{B.3}$$

Together with equation (B.1), these expressions eliminate factor prices from the nonlinear equilibrium problem.

Household optimization gives $P_{os}C_{os} = \beta_s Y_o$ for positive final-demand shares. Zero-share sectors can still supply intermediates. Household disposable income is $Y_o = I_o^L + I_o^K - \sum_s \tau_{os}B_{os} + D_o$. Combining final and intermediate expenditures gives

$$\sum_s E_{os} = \sum_s (1-\tau_{os})B_{os} + D_o = \sum_s R_{os} + D_o \qquad \text{for each } o \in \mathcal{R}.$$

Thus D_o equals provincial expenditure less provincial receipts, with a positive value financing a trade deficit. In the domestic model, $\sum_{o\in\mathcal{R}} D_o = 0$. I express all nominal values, including D_o, in the same numeraire and replace one redundant goods-market equation with the normalization $P^c_{o^*} = 1$ for a reference province o^*.

B.3 Foreign trade in the quantitative model

The quantitative extension described in Section 4.3 adds a foreign region, indexed by 0, that supplies imports and purchases Chinese goods. Let $\mathcal{N} = \mathcal{R} \cup \{0\}$. For Chinese destinations, the origin sums in equations (12) and (15) run over $\mathcal{N}$. Delivered import prices $\kappa^s_{r0} p_{0s}$ are fixed, so foreign supply is perfectly elastic. Foreign purchasers allocate fixed sectoral expenditure $\bar{E}_{0s}$ across Chinese origins according to

$$P_{0s} = \left[\sum_{o\in\mathcal{R}} \left(\kappa^s_{0o} p_{os}\right)^{-\theta_s}\right]^{-1/\theta_s}, \qquad \pi_{0o,s} = \frac{\left(\kappa^s_{0o} p_{os}\right)^{-\theta_s}}{\sum_{m\in\mathcal{R}} \left(\kappa^s_{0m} p_{ms}\right)^{-\theta_s}}. \tag{B.4}$$

The foreign expenditure shares sum to one over Chinese suppliers. With $Q_{0s} = \bar{E}_{0s}/P_{0s}$, demand for the aggregate Chinese sectoral composite has unit price elasticity, while expenditure shifts across Chinese origins according to the sectoral Armington elasticity. Origin receipts include export sales, extending equation (17) to

$$R_{os} = \sum_{r\in\mathcal{R}} \pi_{ro,s} E_{rs} + \pi_{0o,s} \bar{E}_{0s}. \tag{B.5}$$

Domestic production, household accounts, and government objectives retain their definitions in Section 3. National input pools sum only over Chinese provinces, so foreign production does not enter the productivity externality. Provincial expenditure less receipts equals D_o, now accounting for both interprovincial and foreign trade. Summing goods clearing across Chinese provinces gives

$$\sum_{r\in\mathcal{R},s} \pi_{r0,s} E_{rs} - \sum_s \bar{E}_{0s} = \sum_{o\in\mathcal{R}} D_o.$$

National import payments can therefore differ from export receipts. The foreign counterpart is $D_0 = -\sum_{o\in\mathcal{R}} D_o$, so $\sum_{o\in\mathcal{N}} D_o = 0$. Fixed foreign prices anchor the nominal price level, replacing the domestic model's numeraire condition.

B.4 Exact changes

The equations below include the foreign-trade extension. Let b denote the baseline and $\hat{z} = z'/z^b$. Trade imbalances remain fixed in nominal terms in the external numeraire, so

$D'_o = D^b_o$. These amounts can be positive, negative, or zero and enter the system in levels.

To allow for differences in the units of provincial input composites, the quantitative implementation extends the national scale measure in equation (3) to

$$Z_s = \sum_{o \in \mathcal{R}} a_{os} F_{os}, \qquad a_{os} > 0. \tag{B.6}$$

The theoretical specification is the special case $a_{os} = 1$. I set $a_{os} = u^b_{os}/u^{\text{ref}}_s$, where u^{ref}_s is a fixed sectoral reference cost. Since $B^b_{os} = u^b_{os} F^b_{os}$, this calibration makes baseline shares in the national input pool equal to input-bill shares. Equation (B.6) then gives

$$\widehat{Z}_s = \sum_o \chi^b_{os} \frac{\widehat{B}_{os}}{\widehat{u}_{os}}, \qquad \chi^b_{os} = \frac{B^b_{os}}{\sum_r B^b_{rs}}, \tag{B.7}$$

$$\widehat{\Psi}_s = \widehat{Z}^{\phi_s}_s. \tag{B.8}$$

Baseline bills therefore weight changes in physical input use. The conversion coefficients stay fixed, while current pool shares adjust with real inputs. Deflating the bill change by the input-cost change matters because a rise in expenditure need not represent a larger physical input pool. National productivity consistency is imposed within every equilibrium evaluation, including government deviations, so the government anticipates the productivity response when assessing a marginal subsidy. Baseline matching preserves observed productivity levels while allowing policies to change productivity through national scale.

Baseline labor-income and fixed-factor-income shares are

$$\lambda^b_{os} = \frac{\ell_s B^b_{os}}{I^{L,b}_o}, \qquad \eta^b_{os} = \frac{k_s B^b_{os}}{I^{K,b}_o}.$$

Fixed endowments and factor clearing give

$$\widehat{y}_o = \sum_s \lambda^b_{os} \widehat{B}_{os}, \qquad \widehat{r}_o = \sum_s \eta^b_{os} \widehat{B}_{os}, \qquad \widehat{w}_{os} = \widehat{B}^{1/\zeta}_{os} \widehat{y}^{1-1/\zeta}_o. \tag{B.9}$$

The wage expression follows directly by taking ratios in equation (B.3). Input costs and physical inputs satisfy

$$\widehat{u}_{os} = \widehat{r}^{k_s}_o \widehat{w}^{\ell_s}_{os} \prod_j \widehat{P}^{\gamma_{sj}}_{oj}, \qquad \widehat{F}_{os} = \frac{\widehat{B}_{os}}{\widehat{u}_{os}}. \tag{B.10}$$

With fixed fundamental productivity and bilateral trade costs, changes in delivered domestic

prices and destination price indices satisfy

$$\widehat{c}_{os} = \frac{1-\tau'_{os}}{1-\tau^b_{os}}\frac{\widehat{u}_{os}}{\widehat{\Psi}_s}, \qquad \widehat{P}_{rs} = \left(\pi^b_{r0,s} + \sum_{o\in\mathcal{R}} \pi^b_{ro,s}\widehat{c}_{os}^{-\theta_s}\right)^{-1/\theta_s}. \tag{B.11}$$

Here $\widehat{c}_{os}$ is also the factory-price change because $\bar{A}_{os}$ is fixed, and the foreign import price has unit hat. The first expression separates the direct policy wedge from changes in input costs and national productivity. The second gives their effect on destination prices through baseline trade shares.

For a Chinese destination, define

$$\mathcal{D}_{rs} = \pi^b_{r0,s} + \sum_{o\in\mathcal{R}} \pi^b_{ro,s}\widehat{c}_{os}^{-\theta_s}.$$

Then

$$\widehat{P}_{rs} = \mathcal{D}_{rs}^{-1/\theta_s}, \qquad \pi'_{ro,s} = \frac{\pi^b_{ro,s}\widehat{c}_{os}^{-\theta_s}}{\mathcal{D}_{rs}}, \qquad \pi'_{r0,s} = \frac{\pi^b_{r0,s}}{\mathcal{D}_{rs}}. \tag{B.12}$$

For foreign buyers, the denominator is $\mathcal{D}_{0s} = \sum_{o\in\mathcal{R}} \pi^b_{0o,s}\widehat{c}_{os}^{-\theta_s}$, and $\pi'_{0o,s} = \pi^b_{0o,s}\widehat{c}_{os}^{-\theta_s}/\mathcal{D}_{0s}$.

At a candidate bill vector, disposable resources and expenditures are

$$Y'_o = \sum_s (1-\gamma_s-\tau'_{os})B^b_{os}\widehat{B}_{os} + D^b_o, \qquad E'_{rs} = \beta_s Y'_r + \sum_j \gamma_{js}B^b_{rj}\widehat{B}_{rj}.$$

The remaining goods-market equations are

$$(1-\tau'_{os})B^b_{os}\widehat{B}_{os} = \sum_{r\in\mathcal{R}} \pi'_{ro,s}E'_{rs} + \pi'_{0o,s}\bar{E}_{0s}. \tag{B.13}$$

Solving bill and price consistency jointly determines the equilibrium, with national productivity imposed through equation (B.8). The state can be written as log bill hats and log domestic sectoral price hats. Welfare and activity changes are

$$\widehat{P}^c_o = \prod_s \widehat{P}^{\beta_s}_{os}, \qquad \widehat{W}_o = \frac{Y'_o/Y^b_o}{\widehat{P}^c_o}, \qquad \widehat{V}_{os} = \frac{\widehat{B}_{os}}{\widehat{P}^c_o}. \tag{B.14}$$

The equations reproduce the calibrated accounts at unit hats and the baseline policy schedule.

For an exact-change calculation around a baseline equilibrium of the domestic model, omit the foreign-origin and export terms from the trade and goods-market equations and impose $\sum_o D^b_o = 0$. Replace one redundant goods-market equation with $\widehat{P}^c_{o^*} = 1$, holding D_o fixed in this numeraire. The production, factor-market, national-productivity, and welfare equations otherwise retain the same form.

B.5 Input-output parameters and occupational responsiveness

Table B.1 reports the production, demand, trade, and scale parameters. The labor, fixed-factor, and input columns are shares of total production costs and sum to one. The 2007 national input-output table supplies common industry cost shares and final-expenditure shares. Employee compensation determines labor payments. Depreciation and operating surplus determine fixed-factor payments. Net production taxes are excluded from factor-cost shares. Final-demand shares use domestic final expenditure, including consumption, government demand, capital formation, and inventory change, which the model represents as static final absorption. Common shares are held fixed across the three benchmark years, while provincial production and trade patterns vary. The comparison across years therefore incorporates the changing location and scale of activity within a common specification of industry input requirements and final expenditure.

The trade-elasticity assignments follow the Giri–Yi–Yilmazkuday estimates as mapped by Bartelme, Costinot, Donaldson, and Rodríguez-Clare (2025). The unmatched industries are furniture, cultural and arts manufacturing, other manufacturing, waste and recycling, and the 18 nonmanufacturing industries. Their trade elasticity is 4.32, the median of the adopted 15-industry vector, and their EES coefficient is zero. The median pertains to that mapped vector rather than the complete original Giri–Yi–Yilmazkuday sample. Assigned trade elasticities range from 2.97 to 8.94, and Armington substitution elasticities are $1 + \theta_s$. Positive EES elasticities apply to 25 manufacturing industries and range up to 0.22. Separate detailed industries retain distinct national productivity pools.

Table B.1: Sectoral technology and demand parameters

	Sector	β_s	Labor	Fixed factor	Inputs	θ_s	ϕ_s
1	Agriculture	0.0558	0.5565	0.0293	0.4142	4.32	0.00
2	Mining	0.0016	0.1824	0.2408	0.5768	4.32	0.00
3	Food Processing	0.0335	0.0742	0.0780	0.8478	3.57	0.17
4	Food Products	0.0226	0.0642	0.1090	0.8267	3.57	0.17
5	Beverage	0.0084	0.0856	0.1477	0.7666	3.57	0.17
6	Tobacco	0.0082	0.1810	0.1854	0.6336	3.57	0.17
7	Textile	0.0024	0.0765	0.0826	0.8409	4.43	0.14
8	Apparel	0.0162	0.1241	0.0750	0.8009	4.43	0.14
9	Leather	0.0076	0.0887	0.0770	0.8343	4.43	0.14
10	Timber	0.0006	0.0854	0.1021	0.8125	4.32	0.12
11	Furniture	0.0066	0.1195	0.0945	0.7860	4.32	0.00
12	Paper	0.0006	0.0689	0.1035	0.8276	2.97	0.22
13	Printing	0.0002	0.1294	0.1331	0.7375	2.97	0.22
14	Culture Arts	0.0013	0.0897	0.0877	0.8226	4.32	0.00

Table B.1 continued

	Sector	β_s	Labor	Fixed factor	Inputs	θ_s	ϕ_s
15	Petro Processing	0.0028	0.0547	0.0799	0.8654	8.94	0.07
16	Chemical Processing	0.0044	0.0604	0.1009	0.8388	3.75	0.16
17	Pharmaceuticals	0.0046	0.0892	0.1641	0.7468	3.75	0.16
18	Fiber	0.0003	0.0401	0.0969	0.8631	3.75	0.16
19	Rubber Plastic	0.0024	0.0649	0.0993	0.8358	4.13	0.13
20	Non Metal Mineral	0.0014	0.1023	0.1267	0.7710	5.14	0.12
21	Ferrous Metals	0.0019	0.0540	0.1021	0.8439	8.94	0.06
22	Non Ferrous Metals	0.0007	0.0536	0.0896	0.8569	8.94	0.06
23	Metal products	0.0062	0.0737	0.0999	0.8264	5.07	0.12
24	General Machinery	0.0208	0.0856	0.1066	0.8078	3.27	0.19
25	Special Machinery	0.0462	0.0927	0.1011	0.8062	3.27	0.19
26	Transport Equip	0.0524	0.0806	0.0726	0.8468	4.47	0.14
27	Electrical Equip	0.0280	0.0520	0.0880	0.8600	3.27	0.21
28	Communication Equip	0.0263	0.0589	0.0795	0.8616	3.27	0.21
29	Measuring Instruments	0.0054	0.0857	0.0920	0.8223	3.27	0.21
30	Other	0.0093	0.1168	0.0904	0.7928	4.32	0.00
31	Waste Recycling	0.0000	0.0121	0.7956	0.1923	4.32	0.00
32	Utilities	0.0124	0.0748	0.1816	0.7435	4.32	0.00
33	Construction	0.2463	0.1216	0.0871	0.7913	4.32	0.00
34	Wholesale & Retail	0.0401	0.1701	0.3630	0.4670	4.32	0.00
35	Transport, Storage & Post	0.0178	0.1309	0.3065	0.5626	4.32	0.00
36	Accommodation & Catering	0.0237	0.1083	0.2404	0.6513	4.32	0.00
37	Information Services	0.0174	0.1177	0.4683	0.4141	4.32	0.00
38	Finance	0.0182	0.1942	0.4690	0.3368	4.32	0.00
39	Real Estate	0.0452	0.1039	0.7055	0.1906	4.32	0.00
40	Leasing & Business Services	0.0074	0.1157	0.1839	0.7004	4.32	0.00
41	Scientific & Technical Services	0.0073	0.2871	0.2064	0.5065	4.32	0.00
42	Water, Environment & Public Facilities	0.0059	0.2603	0.2442	0.4955	4.32	0.00
43	Resident & Other Services	0.0165	0.1343	0.3077	0.5580	4.32	0.00
44	Education	0.0484	0.4453	0.1078	0.4469	4.32	0.00
45	Health & Social Work	0.0430	0.2327	0.1026	0.6648	4.32	0.00
46	Culture, Sports & Entertainment	0.0069	0.2055	0.1958	0.5988	4.32	0.00
47	Public Administration & Social Security	0.0645	0.4775	0.0703	0.4522	4.32	0.00

Occupational parameter. The auxiliary calibration maps 2008 employment and wage bills to broader input-output industries. A source category split across destination industries is allocated equally among them. With observed employment share $\widetilde{\lambda}_{os}$ and average earnings

$\bar{e}_{os}$, the employment-weighted regression is

$$\log \widetilde{\lambda}_{os} = \delta_o + b \log \bar{e}_{os} + \varepsilon_{os}.$$

It yields $b = 1.391151$ across 909 cells in 31 provinces, including Tibet, and the model sets $\zeta = b$. This auxiliary sample differs from the 30-province quantitative sample. Since the maintained Roy model equalizes conditional mean realized earnings across occupations, the regression slope is an auxiliary responsiveness benchmark rather than a structurally identified estimate of the efficiency-wage elasticity. The same value is held fixed in each model year.

B.6 Reconciliation of bilateral flows

The 42-sector interregional input-output tables provide priors for the model's 47-sector flows. A detailed model industry inherits its parent industry's provincial routing profile. Combined source industries contribute their summed flows. National export-to-output and import-to-use ratios determine initial foreign-trade totals, which are allocated across provinces using those profiles.

I measure provincial trade imbalances from the raw interregional input-output accounts before mapping industries. Let D_o^{IRIO} be purchases from other provinces and abroad, including final and intermediate goods, less sales to those destinations, and let GO_o^{IRIO} be provincial gross output in the same accounts. I preserve the ratio of net imports to gross output when expressing the imbalance in benchmark units:

$$D_o^b = \frac{D_o^{\text{IRIO}}}{GO_o^{\text{IRIO}}} \sum_s R_{os}^b. \tag{B.15}$$

This scaling accommodates the difference between nominal input-output accounts and the constructed benchmark output series. Positive values finance deficits and negative values correspond to surpluses. The construction permits a national trade imbalance, with the external counterpart equal to $-\sum_o D_o^b$.[20]

Given these imbalances, benchmark household resources are $Y_o^b = \sum_s (1-\gamma_s-\tau_{os}^b) B_{os}^b + D_o^b$, and sectoral expenditure follows equation (16). I hold observed producer receipts and input bills fixed and adjust bilateral flows to match these expenditure and receipt margins. The resulting flows preserve each province's calibrated net imports while satisfying the model's accounts.

For each industry and year, let $Q_{ro,s}$ denote a prior flow from origin o to destination r. I

[20]The 2012 and 2017 source tables include Tibet, which is outside the model's 30-province sample. Its bilateral flows are included in the external counterpart when measuring the retained provinces' net imports.

choose calibrated flows by

$$\begin{aligned} \min_{f_{ro,s}\geq 0} \quad & \sum_{r,o:Q_{ro,s}>0} [f_{ro,s}\log(f_{ro,s}/Q_{ro,s}) - f_{ro,s} + Q_{ro,s}] \\ \text{subject to} \quad & \sum_{o\in\mathcal{N}} f_{ro,s} = E_{rs} \quad (r\in\mathcal{R}), \\ & \sum_{r\in\mathcal{N}} f_{ro,s} = R_{os} \quad (o\in\mathcal{R}). \end{aligned} \tag{B.16}$$

The rest-of-world margins adjust, and foreign trade with itself is zero. Structural zeros remain zero. Economically available links with zero source flows receive a small positive prior. On supported positive flows the solution has the form $f_{ro,s} = a_r Q_{ro,s} b_o$, with multipliers chosen by iterative proportional fitting to satisfy both margins. These calibration multipliers are unrelated to the national-input conversion coefficients.

Domestic trade shares are $\pi_{ro,s} = f_{ro,s}/E_{rs}$. Foreign spending and its origin shares are $\bar{E}_{0s} = \sum_o f_{0o,s}$ and $\pi_{0o,s} = f_{0o,s}/\bar{E}_{0s}$. The reconciliation supplies expenditure shares and accounts consistent with the specified trade imbalances for the exact-change system. It does not separately identify the levels of bilateral trade costs and fundamental productivities. The quantitative comparisons retain these baseline shares and fixed bilateral trade costs.

C Priority Recovery and Provincial Optimality

The inverse problem uses the response of the complete competitive equilibrium to unilateral policy changes. Rival policy rates and trade imbalances are fixed, while goods prices, factor allocation, fiscal resources, and national productivity adjust. This appendix derives the response system and separates recovery of first-order incentives from verification of provincial optimality. The response equations include the foreign-trade extension in Appendix B.3 and the national input aggregate in equation (B.6). The domestic model in Section 3 follows by setting $a_{os} = 1$, omitting the foreign-trade terms, and imposing its price normalization; the local recovery argument applies to both specifications.

C.1 Differentiating the competitive equilibrium

Write the equilibrium in Appendix B as $\mathcal{E}(z,\tau) = 0$, where z collects log input bills and sectoral prices. For the domestic model, the system includes the numeraire condition in place of one redundant goods-market equation. The national productivity equations are included

by substitution. On a branch with nonsingular $\mathcal{E}_z$, the response to a policy direction d is

$$z_d = -\mathcal{E}_z^{-1}\mathcal{E}_\tau d. \tag{C.1}$$

The factorization of $\mathcal{E}_z$ can be reused across policy directions. The following equations describe its economic components explicitly.

Consider $\tau(t) = \tau + td$. Denote derivatives of log bills, input costs, factory prices, and composite prices by b_{os}, $u_{os}^{(1)}$, $c_{os}^{(1)}$, and $p_{os}^{(1)}$. These are response variables rather than levels. Let λ_{os} and η_{os} be current labor-income and fixed-factor-income shares. Factor clearing and input costs give

$$\begin{aligned}
(\log y_o)' &= \sum_s \lambda_{os} b_{os}, \qquad (\log r_o)' = \sum_s \eta_{os} b_{os}, \\
(\log w_{os})' &= \zeta^{-1} b_{os} + (1-\zeta^{-1}) \sum_j \lambda_{oj} b_{oj}, \\
u_{os}^{(1)} &= k_s (\log r_o)' + \ell_s (\log w_{os})' + \sum_j \gamma_{sj} p_{oj}^{(1)}.
\end{aligned} \tag{C.2}$$

With current physical-pool shares $\chi_{os} = a_{os}F_{os}/Z_s$ from equation (B.6),

$$\begin{aligned}
(\log \Psi_s)' &= \phi_s \sum_o \chi_{os}(b_{os} - u_{os}^{(1)}), \\
c_{os}^{(1)} &= u_{os}^{(1)} - (\log \Psi_s)' - \frac{d_{os}}{1-\tau_{os}}, \\
p_{rs}^{(1)} &= \sum_{o \in \mathcal{R}} \pi_{ro,s} c_{os}^{(1)}, \qquad (\log \pi_{ro,s})' = -\theta_s (c_{os}^{(1)} - p_{rs}^{(1)}).
\end{aligned} \tag{C.3}$$

Foreign import prices have zero response. For foreign demand, replace the destination index by 0 and form $p_{0s}^{(1)}$ using export shares across Chinese origins.

Disposable income is $Y_o = \sum_s (1-\gamma_s-\tau_{os})B_{os} + D_o$. Holding D_o fixed in the relevant numeraire, income and expenditure responses satisfy

$$\begin{aligned}
Y_o' &= \sum_s (1-\gamma_s-\tau_{os}) B_{os} b_{os} - \sum_s B_{os} d_{os}, \\
E_{rs}' &= \beta_s Y_r' + \sum_j \gamma_{js} B_{rj} b_{rj}.
\end{aligned} \tag{C.4}$$

Differentiating goods clearing gives

$$(1-\tau_{os})B_{os}b_{os} - B_{os}d_{os} = \sum_r \pi_{ro,s} E_{rs} \left[(\log \pi_{ro,s})' + \frac{E_{rs}'}{E_{rs}} \right] + \pi_{0o,s} \bar{E}_{0s} (\log \pi_{0o,s})'. \tag{C.5}$$

Equations (C.2)–(C.5) form a linear system for equilibrium responses. They incorporate the direct fiscal cost of policy, changes in subsidy bases, and national productivity feedback. Welfare and activity derivatives follow from

$$(\log W_o)' = Y_o'/Y_o - \sum_s \beta_s p_{os}^{(1)}, \qquad (\log V_{os})' = b_{os} - \sum_j \beta_j p_{oj}^{(1)}.$$

Applying these equations to the individual own-policy directions constructs a_o and $\mathcal{B}_o$ in equation (24).

C.2 Recovery and local implementation

Proposition 1 (Recovery and local support)**.** *Fix an interior subsidy profile and a twice continuously differentiable competitive-equilibrium branch. Suppose province o has the same number of controlled rates and valued activities, and its activity-response matrix $\mathcal{B}_o$ is nonsingular. The unique priorities satisfying its first-order conditions are $\boldsymbol{\vartheta}_o = -\mathcal{B}_o^{-1} a_o$. If the resulting own-policy Hessian H_o in equation (26) is negative definite, the provincial policy is a strict local best response. If these conditions hold for every province, the profile has strict local Nash support.*

Proof. Differentiating the government objective yields $a_o + \mathcal{B}_o \boldsymbol{\vartheta}_o$. Nonsingularity gives the unique zero of this expression in priorities. With those priorities fixed, a unilateral policy perturbation d_o changes the objective by

$$g_o(\tau_o + t d_o, \tau_{-o}) - g_o(\tau) = \frac{t^2}{2} d_o^{\mathsf{T}} H_o d_o + o(t^2).$$

Negative definiteness and continuity yield strict local optimality in all own-policy directions. Applying the argument to each province establishes the local Nash property. □

The activity-coordinate representation makes the result particularly transparent. Let v_o collect the log sectoral activities corresponding to the columns of $\mathcal{B}_o$, and let $J_o = D_{\tau_o} v_o = \mathcal{B}_o^{\mathsf{T}}$. A local inverse expresses welfare as $f_o(v_o) = \log W_o(\tau_o(v_o), \tau_{-o})$. The government objective becomes $f_o(v_o) + \boldsymbol{\vartheta}_o^{\mathsf{T}} v_o$. At the recovered first-order conditions,

$$\boldsymbol{\vartheta}_o = -\nabla_v f_o, \qquad H_o = J_o^{\mathsf{T}} D_v^2 f_o J_o. \tag{C.6}$$

The terms involving second derivatives of the coordinate map vanish because the gradient of the objective is zero. Thus identification uses the rank of the activity response, whereas local optimality uses the curvature of resident welfare in those activity coordinates.

At bounds $\underline{\tau}_{ok} \le \tau_{ok} \le \overline{\tau}_{ok}$, write $q_o = a_o + \mathcal{B}_o \boldsymbol{\vartheta}_o$. The necessary signs for a maximizer are

$$(q_o)_k \le 0 \text{ at a lower bound}, \qquad (q_o)_k = 0 \text{ in the interior}, \qquad (q_o)_k \ge 0 \text{ at an upper bound}. \tag{C.7}$$

Second-order sufficiency requires negative curvature on feasible directions with zero first-order gain. Historical recovery treats observed signed rates, including zero rates, as interior choices. The nonnegative counterfactual domain instead treats zero as a lower bound.

C.3 Second derivatives and economic curvature

For two policy directions d, e, implicit differentiation gives

$$\mathcal{E}_z z_{de} = -\left\{\mathcal{E}_{zz}[z_d, z_e] + \mathcal{E}_{z\tau}[z_d, e] + \mathcal{E}_{\tau z}[d, z_e] + \mathcal{E}_{\tau\tau}[d, e]\right\}. \tag{C.8}$$

Second responses therefore use the same equilibrium Jacobian as first responses. For a scalar objective $g(z, \tau)$, a transpose solve $\mathcal{E}_z^{\mathsf{T}} \lambda = g_z$ projects the second-order forcing directly into objective curvature. This adjoint calculation avoids a separate state solve for every outcome.

The nonlinear forcing terms have explicit economic forms. For a weighted log sum, the second derivative equals the weighted mean of component second derivatives plus the variance of component first derivatives. In particular,

$$(\log Z_s)'' = \sum_o \chi_{os} (\log F_{os})'' + \operatorname{Var}_{\chi_s}\left((\log F_{os})'\right), \tag{C.9}$$

$$(\log P_{rs})'' = \sum_o \pi_{ro,s} (\log p_{os})'' - \theta_s \operatorname{Var}_{\pi_{r\cdot,s}}\left((\log p_{os})'\right). \tag{C.10}$$

The foreign import component has zero derivatives. Factor-income aggregates contribute analogous positive variance terms. Along a straight path in level subsidy rates, $(\log(1-\tau_{os}))'' = -d_{os}^2/(1-\tau_{os})^2$. The fiscal second derivative retains both bill curvature and the interaction between the policy change and its changing base. These terms are included before assembling the complete own-policy Hessian.

The economic sources of curvature can also be seen from the provincial resource identity. Define real sales revenue $\mathcal{R}_o = \sum_s R_{os}/P_o^c$ and real intermediate expenditure $\mathcal{M}_o = \sum_s \gamma_s B_{os}/P_o^c$. Then $W_o = \mathcal{R}_o - \mathcal{M}_o + D_o/P_o^c$. Let $r = \log \mathcal{R}_o$, $h = \log \mathcal{M}_o$, and $p = \log P_o^c$. Holding nominal D_o fixed, the Hessian in activity coordinates is

$$\begin{aligned} D_v^2 \log W_o = {} & \frac{\mathcal{R}_o}{W_o}\left[D_v^2 r + (D_v r)(D_v r)^{\mathsf{T}}\right] - \frac{\mathcal{M}_o}{W_o}\left[D_v^2 h + (D_v h)(D_v h)^{\mathsf{T}}\right] \\ & + \frac{D_o}{P_o^c W_o}\left[(D_v p)(D_v p)^{\mathsf{T}} - D_v^2 p\right] - (D_v \log W_o)(D_v \log W_o)^{\mathsf{T}}. \end{aligned} \tag{C.11}$$

This expression permits positive, negative, or zero D_o, provided $W_o > 0$. When $D_o = 0$, it reduces to

$$D_v^2 \log W_o = \frac{D_v^2 r - m D_v^2 h}{1-m} - \frac{m}{(1-m)^2}(D_v r - D_v h)(D_v r - D_v h)^{\mathsf{T}}, \qquad m = \frac{\mathcal{M}_o}{\mathcal{R}_o}.$$

When all activities are coordinates, intermediate expenditure is a sum of exponentials of those activities and its log Hessian is a positive semidefinite expenditure-share covariance. In the balanced-trade case, its negative contribution and the final outer-product term provide stabilizing curvature. With nonzero D_o, changes in the consumption price index also change the real value of the inflow; the general expression retains this additional source of curvature. In the application, four background industries also respond endogenously, and their second derivatives enter $D_v^2 h$. The calculation retains these responses. Receipt curvature can have either sign, so private constant returns alone does not establish government concavity.

C.4 Recovery diagnostics

Table C.1 reports the diagnostics for observed policies. Maxima are taken across the 30 provincial systems in each year. First-order residuals are derivatives of the log government objective with respect to subsidy rates. All provincial response matrices are full rank, with maximum condition numbers below 8.2. First-order residuals are at numerical precision. The largest eigenvalue of every full own-policy Hessian is strictly negative. These checks evaluate the revealed priorities held fixed while the province changes its policies, and they include all cross-sector own-policy terms.

Table C.1: Priority recovery and joint own-policy curvature

Year	Controls	Full rank	Max. condition	Max. FOC	Largest eigenvalue
2007	1,288	30	8.2	5.55×10^{-16}	-8.09×10^{-6}
2012	1,285	30	7.6	1.11×10^{-16}	-2.17×10^{-5}
2017	1,286	30	7.1	4.44×10^{-16}	-4.39×10^{-5}

Table C.2 reports forward policy calculations from four starting profiles in each benchmark year: zero rates in all controlled cells, a uniform rate within each province, and two independent randomized subsidy schedules. With the recovered priorities held fixed, all twelve calculations return to the observed subsidy profile within 10^{-6} in rate levels. Convergence takes five to seven rounds of sequential provincial adjustment, and every endpoint passes the first- and second-order checks for all thirty provincial objectives. Together, these results provide numerical support for the observed subsidy profile as the unique Nash equilibrium under the

recovered priorities.

I also repeat the searches with each province maximizing resident welfare under the non-negative subsidy bounds used in the priority-removal counterfactual. All twelve calculations pass the provincial first- and second-order checks. The largest pairwise differences in final subsidy rates are 6.16×10^{-5}, 4.20×10^{-5}, and 2.92×10^{-6} in 2007, 2012, and 2017. The national welfare spread across starting profiles is at most 6.69×10^{-8} percentage points, and the largest provincial welfare difference from the reported welfare-only outcome is 2.78×10^{-7} percentage points. These comparisons show that the reported welfare-only outcomes are stable across starting profiles.

Table C.2: Provincial policy games from alternative starting profiles

	2007	2012	2017
Panel A: Recovered priorities			
Starts returning to observed rates	4/4	4/4	4/4
Largest rate difference from observed	6.07×10^{-7}	9.89×10^{-7}	2.22×10^{-7}
Rounds of provincial adjustment	5–6	5–6	6–7
Panel B: Welfare-only objectives			
Converged starting profiles	4/4	4/4	4/4
Largest pairwise rate difference	6.16×10^{-5}	4.20×10^{-5}	2.92×10^{-6}
National welfare spread across starts (p.p.)	1.84×10^{-8}	6.69×10^{-8}	2.44×10^{-8}
Max. provincial welfare difference (p.p.)	1.06×10^{-7}	2.78×10^{-7}	5.52×10^{-8}

The plan and growth comparisons in Section 4.5 provide complementary evidence about the revealed objectives.

C.5 Responses to changes in priorities

To distinguish historical recovery from forward policy changes, stack governments' first-order conditions as $K(\tau, \vartheta) = 0$. On a regular interior policy-equilibrium branch,

$$\frac{d\tau}{dt} = -K_\tau^{-1} K_\vartheta \frac{d\vartheta}{dt}. \tag{C.12}$$

The diagonal blocks of K_τ are own-policy Hessians. The off-diagonal blocks describe how rival policies affect marginal incentives. A change in one province's priorities can therefore induce adjustments throughout the policy profile. At bounds, the corresponding system applies to free coordinates with the relevant active constraints. Finite policy experiments solve these conditions jointly with competitive equilibrium and check provincial optimality at the resulting policies.

D Counterfactual Construction and Outcome Measures

The policy experiments in this appendix use the quantitative model with foreign trade described in Appendix B.3.

D.1 Policy experiments and outcome measures

The all-zero-policy experiment sets every effective rate in the 47-sector economy to zero and solves competitive equilibrium jointly with national productivity. The priority-removal experiment instead replaces each provincial objective by $\log W_o$ and optimizes rates in the active strategic sectors over $[0, 0.95]$. Public-service and absent-production policy cells remain at observed values. The first experiment removes subsidies directly; the second changes government objectives and solves for the resulting subsidies. Zero priorities do not impose zero support.

Appendix G.6 repeats these experiments under alternative calibrations. Appendix G.7 derives the welfare decomposition and reports its integrated contributions.

National welfare is the fixed-weight geometric index in equation (29). For a sector group $\mathcal{A}$, aggregate real-value-added changes are

$$100\left[\frac{\sum_o \sum_{s\in\mathcal{A}}(1-\gamma_s)B^b_{os}\widehat{B}_{os}/\widehat{P}^c_o}{\sum_o \sum_{s\in\mathcal{A}}(1-\gamma_s)B^b_{os}} - 1\right]. \tag{D.1}$$

Amounts are summed before calculating the percentage change. The manufacturing group contains 29 sectors. The nonmanufacturing activity group contains the other 18, including the four public services. Rate averages instead use the 14 strategic nonmanufacturing categories and fixed observed input-bill weights. Spending shares use the policy rates and input bills of the allocation being described.

Let $v_o = (V_{os})_{s\in\mathcal{M}}$ be province o's vector over the manufacturing sectors. Manufacturing similarity is

$$C = \frac{1}{209}\sum_{(o,r)\in\mathcal{P}_{IC}} \frac{v_o^\top v_r}{\|v_o\|\|v_r\|}, \tag{D.2}$$

where $\mathcal{P}_{IC}$ contains the 209 pairs formed from 19 inland and 11 coastal provinces. A province-common consumption deflator cancels from each cosine. The main zero-policy table reports a percentage change in C. Table D.1 reports changes in cosine points. The five-year magnitude comparison divides $C^{\text{cf}}_t - C^b_t$ by $(C^b_{t+5} - C^b_t)/5$, using the same quantitative data at both dates.

Rates within 10^{-7} of zero are counted as zero in Table D.1.

The 2012 observed strategic profile is feasible under nonnegative policies. For 2007 and

Table D.1: Welfare-only provincial policy across calibrated years

	2007	2012	2017
National welfare (percent)	−0.2698	−0.0681	−0.1168
Aggregate real value added (percent)	−0.9430	+0.2625	−0.4029
Manufacturing cosine change (points)	+0.003512	+0.004157	−0.000458
Provinces gaining in welfare	4/30	8/30	7/30
Strategic rates at zero	917/1,288	934/1,285	994/1,286

2017, I first solve the nonnegative game with the original historical priorities to distinguish the domain restriction from the change in objectives. Relative to observed allocations, national welfare changes by −0.000098 and −0.000405 percent. Starting from these feasible historical-priority profiles, removing priorities changes welfare by −0.269720 and −0.116347 percent, respectively. The small domain adjustment therefore leaves the economic conclusions unchanged. These comparisons use ratios of welfare indices rather than subtracting percentage changes.

Appendix Table C.2 reports the forward policy-adjustment checks under recovered priorities and welfare-only objectives. The welfare-only searches closely agree on national welfare even where policy differences from the reported welfare-only policy profile exceed the policy-comparison tolerance.

E Priority Design and Fiscal Capacity

E.1 Computing constrained-efficient subsidies

The 2012 national policy problem uses the observed factor-income weights, the same subsidy instruments as in the main text, and nonnegative rate bounds. Let $F_o(\boldsymbol{\tau})$ be real provincial expenditure as defined in equation (31). For a multiplier k, the fiscal constraint is $F_o(\boldsymbol{\tau}) \leq kF_o(\boldsymbol{\tau}^b)$. The main specification sets $k = 1$. A second sets $k = 1.25$.

The constraint is on real net expenditure across all sectors. With nonnegative subsidies and the retained 2012 background policies, net and gross support coincide. For reporting, national real expenditure sums provincial amounts in baseline consumption units, with each baseline consumption-price hat normalized to one. The ceilings remain tied to observed spending throughout the calculation. At both finite-cap solutions, all 30 fiscal constraints bind. The computed schedules satisfy feasibility, positive household resources, equilibrium consistency, and the national first-order conditions, including the conditions at subsidy bounds.

I also verify the constrained second-order sufficient conditions at both allocations. The Hessian of the national Lagrangian in equation (E.1) includes the welfare effects of joint policy changes across provinces and the curvature of every real-spending constraint. Its projection onto feasible critical directions is negative definite at both computed allocations, including conservative checks that release nearly weak rate bounds. These checks establish strict local optimality to numerical precision.

For the observed-budget problem, I additionally repeat the optimization from zero subsidies, province-uniform rates, and two randomized sectoral schedules, using tighter stopping criteria. All four searches satisfy the first-order and fiscal conditions. Their maximum pairwise rate difference is below 2.0×10^{-6}, and each reproduces the reported national welfare gain to within 1.7×10^{-8} percentage point. Relative to the reported schedule, the largest rate difference is 4.0×10^{-4}, exceeding the 10^{-4} policy-comparison tolerance. The tighter searches closely agree with one another and reproduce the welfare gain despite these differences in some policy rates. These are numerical checks of solution stability, not a global-optimality or uniqueness proof.

Appendix G.6 reports the redesign under alternative calibrations. Each problem uses the corresponding calibration's observed provincial real-spending ceilings and recovers redesigned priorities at its resulting allocation.

Changes in Table E.1 are relative to the observed 2012 economy.

Table E.1: Priority design and provincial fiscal capacity, 2012

	Observed real budgets	Real budgets plus 25 percent
National welfare (percent)	1.276	1.564
Aggregate real value added (percent)	1.091	1.874
Real spending (percent change)	0.0	25.0
Real spending / baseline factor VA (percent)	2.104	2.630
Provinces gaining in welfare	28/30	29/30
Smallest provincial welfare change (percent)	−0.055	−0.033
Manufacturing cosine	0.5786	0.5802

Expanding each provincial real budget by 25 percent raises the national welfare gain from 1.276 to 1.564 percent. The fixed-budget redesign therefore achieves about 82 percent of the expanded-budget gain. Twenty-eight provinces gain under the fixed-budget reform and 29 under the expanded-budget reform; neither comparison imposes provincial no-loss constraints.

E.2 Fiscal shadow values and redesigned priorities

The national Lagrangian is

$$\mathcal{L}^N = \sum_r \omega_r \log W_r - \sum_r \lambda_r (F_r - \overline{F}_r), \qquad \lambda_r \geq 0, \tag{E.1}$$

with the rate bounds imposed separately. Write $\eta_o^N, \xi_o^N \geq 0$ for the national lower- and upper-rate-bound multipliers and $\eta_o^P, \xi_o^P \geq 0$ for their provincial counterparts. National stationarity with respect to province o's rates is

$$\omega_o a_o + h_o - \sum_r \lambda_r \nabla_{\tau_o} F_r + \eta_o^N - \xi_o^N = 0, \tag{E.2}$$

where $h_o = \sum_{r \neq o} \omega_r \nabla_{\tau_o} \log W_r$. Provincial stationarity is

$$a_o + \mathcal{B}_o \boldsymbol{\vartheta}_o + \eta_o^P - \xi_o^P = 0. \tag{E.3}$$

Subtracting the weighted provincial equation from the national condition gives

$$\omega_o \mathcal{B}_o \boldsymbol{\vartheta}_o = h_o - \sum_r \lambda_r \nabla_{\tau_o} F_r + \eta_o^N - \xi_o^N - \omega_o \eta_o^P + \omega_o \xi_o^P. \tag{E.4}$$

The bound multipliers are complementary to their respective slacks. On coordinates interior to both domains, their terms vanish and this reduces to equation (33). At a lower rate bound, a maximizer's derivative can be negative. The nonnegative lower-bound multiplier offsets it in the displayed stationarity convention.

Real expenditure responds through its rate, spending base, and deflator:

$$\mathrm{d}F_r = \frac{1}{P_r^c} \sum_s \left(B_{rs}\, \mathrm{d}\tau_{rs} + \tau_{rs}\, \mathrm{d}B_{rs} \right) - F_r\, \mathrm{d} \log P_r^c. \tag{E.5}$$

The direct rate term is zero for a change in another province's policy, but the base and price responses generally remain. Thus the fiscal adjustment in equation (E.4) includes every provincial constraint. The term h_o includes all welfare spillovers through national productivity, trade, prices, and financing, rather than only the technological externality.

For a smooth local solution with a stable active set, the envelope theorem gives $\mathrm{d}G^* = \sum_r \lambda_r\, \mathrm{d}\overline{F}_r$. If the constraints are normalized as $F_r / F_r(\boldsymbol{\tau}^b) \leq k$ with multipliers μ_r, then $\lambda_r = \mu_r / F_r(\boldsymbol{\tau}^b)$ and $\mathrm{d}G^*/\mathrm{d}k = \sum_r \mu_r$. The finite comparison between $k = 1$ and $k = 1.25$ allows all policies and equilibrium quantities to adjust. It is not a linear extrapolation using a single set of shadow values.

E.3 Implementation checks

For each constrained-efficient equilibrium, I solve the inverse derivative system for one redesigned priority vector per province. The selected assignment sets the provincial objective gradient to zero. Priorities remain unrestricted in sign and retain a unit coefficient on log resident welfare.

The provincial implementation game uses the original nonnegative rate domain and local financing rule. The national real-spending constraints are not imposed on unilateral deviations. Full own-policy second-order checks establish strict local optimality for all 30 provinces at both constrained-efficient allocations. The same priority vectors are then used for finite searches and forward policy adjustment.

At each allocation, I also evaluate 300 feasible finite policy deviations across provinces, including directions selected from the own-policy curvature. None yields an objective gain above the 10^{-8} tolerance; the largest recorded gain is below 6.4×10^{-13}. Together with the full own-policy second-order checks, these calculations provide numerical evidence supporting the redesigned priorities. Finite searches do not establish global best responses.

For the observed-budget redesign, forward sequential provincial adjustment is also run from the observed policy profile and from a perturbed constrained-efficient subsidy schedule. Both return to a neighborhood of the constrained-efficient schedule. Final response checks use the common resulting profile, and the same redesigned priority vectors are used throughout. These calculations provide numerical evidence of implementation under the specified adjustment procedure; they do not establish equilibrium uniqueness.

E.4 Production-network summaries

The upstreamness comparison uses the full domestic production network. Let Z^D_{sj} be domestic intermediate sales by supplier sector s to user sector j, and let R_s denote national sectoral receipts. Define $Q_{sj} = Z^D_{sj}/R_s$. Domestic upstreamness solves

$$U_s = 1 + \sum_j Q_{sj} U_j, \qquad U = (I - Q)^{-1}\mathbf{1}. \tag{E.6}$$

Domestic final sales and exports are terminal demand. The underlying calculation uses 2012 flows and the common production coefficients, and reconciles receipts with domestic intermediate sales, domestic final sales, and exports. It measures distance to final use in the domestic network, rather than global upstreamness. The main figure compares 29 manufacturing industries, using equal-sector descriptive associations.

The Pearson correlation of mean priority changes with upstreamness is 0.612. The

Spearman correlation is 0.570. Omitting chemicals and nonferrous metals together leaves the Pearson correlation at 0.568. The positive association therefore extends beyond the two largest mean increases. These comparisons organize the sectoral pattern of the jointly designed allocation, without identifying separate causal effects of network position.

Appendix Table G.9 repeats the comparison under alternative calibrations. Each priority change is measured relative to that calibration's recovered priorities, and upstreamness is computed from its own observed production network using (E.6).

F Additional Evidence on Industrial Composition and Priorities

F.1 Additional measures of industrial composition

The Krugman specialization index compares each province's industrial composition with that of the rest of China. Using the manufacturing shares defined in equation (A.2), the index for province o in period p is

$$K_{o,p} = \frac{1}{2} \sum_{s \in \mathcal{M}} \left| x_{os,p} - x_{-o,s,p} \right|, \tag{F.1}$$

where $x_{-o,s,p}$ is sector s's share in pooled constructed real value added across all provinces other than o. The index lies between zero and one. Lower values indicate a manufacturing composition closer to that of the rest of the country.

Figure F.1 shows that inland specialization falls after 2006–08. The inland mean declines from 0.379 to 0.331 by 2015–18, while the coastal mean rises from 0.286 to 0.299. The gap between the two regional means narrows by 65 percent. The specialization index thus confirms the increasing similarity in industrial composition documented by the cosine measure in Section 2.

Figure F.2 shows the mean manufacturing cosine across all 435 distinct province pairs. It rises from 0.575 in 2006–08 to 0.610 in 2015–18. This broader comparison complements the fixed-coastal-benchmark measure by averaging bilateral similarity throughout the country.

Table F.1 separates this change into comparisons within and between regions. The inland–coastal mean rises by 0.073 between 2006–08 and 2015–18, while mean similarity within either region changes little. Thus, the increase in average bilateral similarity mainly reflects the narrowing of differences between inland and coastal provinces.

Employment shares also become more similar across inland and coastal provinces after 2006–08. Figure F.3 reports pairwise cosines using year-end urban-unit employment from the *China Labour Statistical Yearbook* (National Bureau of Statistics of China 2001–2023).



Figure F.1: Manufacturing specialization by region

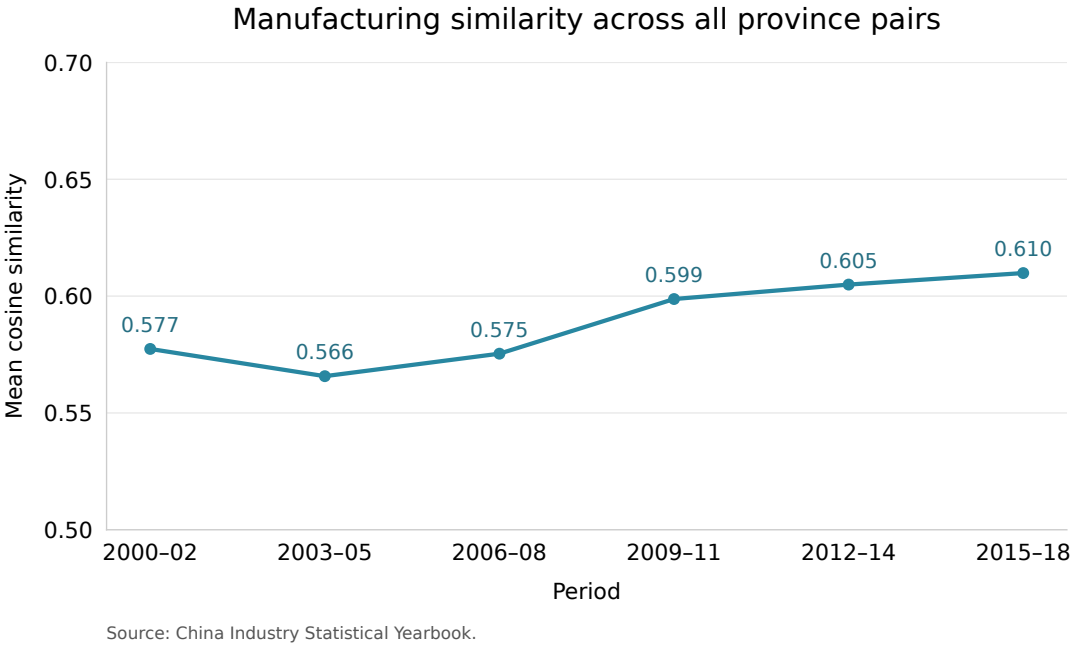


Figure F.2: Manufacturing similarity across all province pairs

Table F.1: Manufacturing similarity within and between regions

Period	Inland–coastal	Inland–inland	Coastal–coastal
2006–08	0.521	0.615	0.659
2009–11	0.559	0.623	0.674
2012–14	0.579	0.616	0.669
2015–18	0.594	0.615	0.654

The comparison covers the same 30 provinces and 29 manufacturing industries, with an extension through 2019–21. The 2015–18 employment period includes all four years, while the constructed-value-added series excludes 2017. The increase in inland–coastal similarity is smaller than for constructed value added, and similarity within the inland group declines. The employment evidence therefore supports the direction of the cross-region change, while showing that its magnitude and within-region pattern depend on the activity measure.

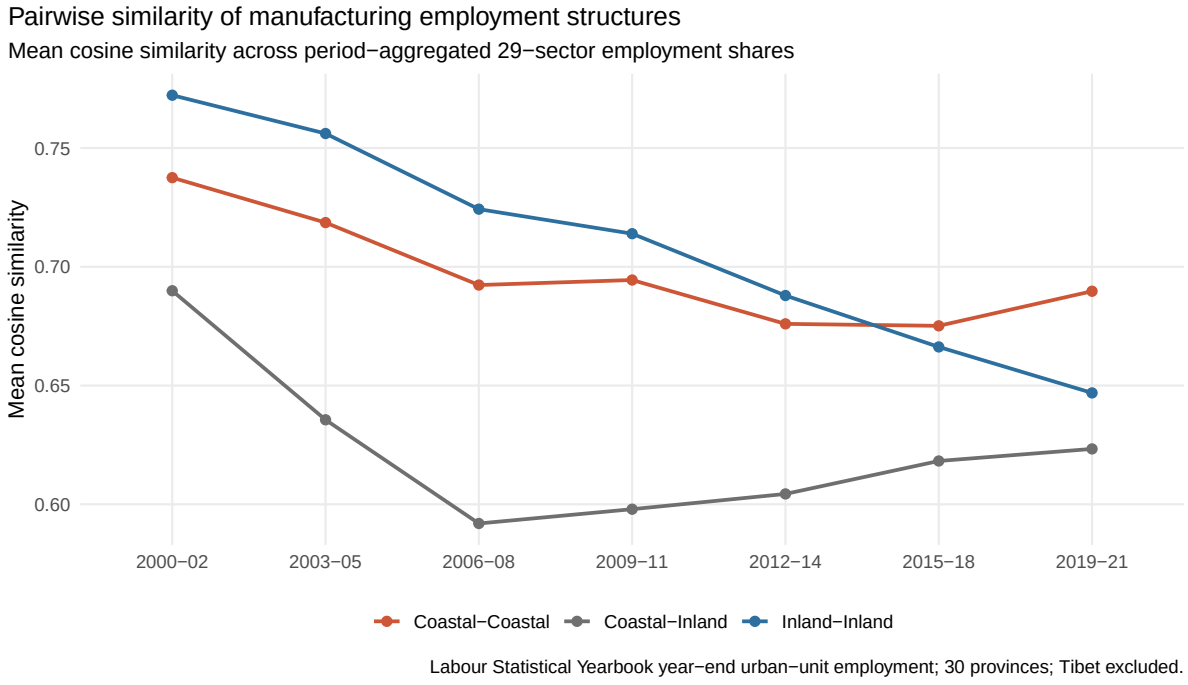
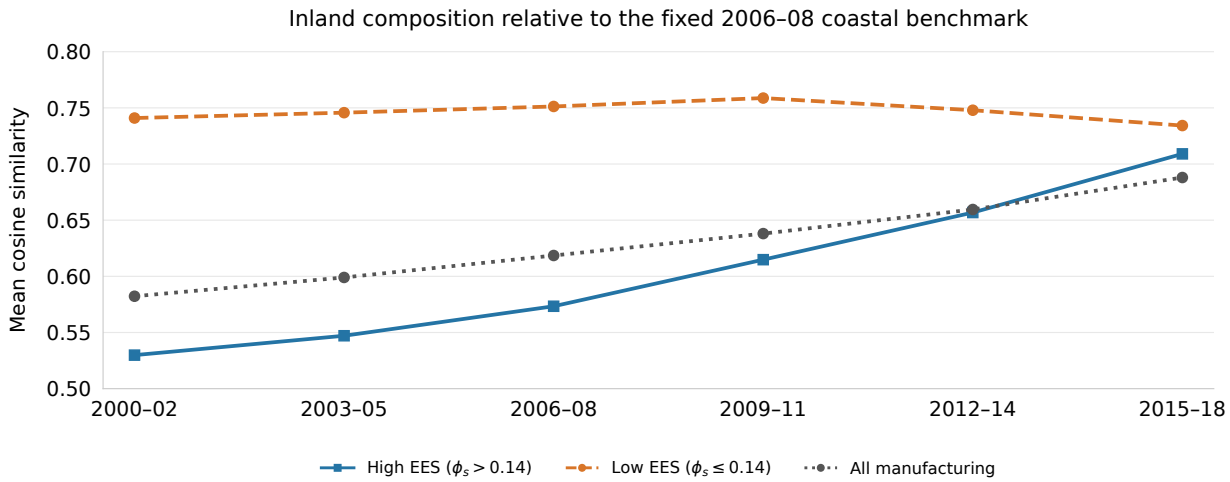


Figure F.3: Manufacturing employment similarity within and between regions

The change in similarity is stronger among industries with larger calibrated external economies of scale. Figure F.4 reports cosines computed separately within two groups of manufacturing industries. Among industries above the median scale elasticity, the inland mean cosine to the fixed coastal benchmark rises from 0.573 in 2006–08 to 0.709 in 2015–18. The corresponding low-elasticity mean changes from 0.751 to 0.734. The comparison

describes the composition of the observed change using the scale elasticities adopted in the quantitative model. ICT accounts for a large share of the aggregate increase in similarity, so this group comparison should not be interpreted as an independent effect of scale elasticities on convergence.



Source: China Industry Statistical Yearbook.

Figure F.4: Inland similarity by manufacturing scale elasticity

F.2 Priority comparisons by industry coverage

Table F.2 distinguishes the main 43-sector comparison from its manufacturing subset. The broad-sector increase in inland–coastal alignment occurs mainly in 2007–12, while the manufacturing comparison continues to rise through 2017. The set of industries therefore matters for the timing of increasing alignment in provincial priorities. Both calculations use signed priorities with the sector set held fixed within each pair across all three dates. Each pair retains at least 25 manufacturing industries.

Table F.2: Inland–coastal priority similarity by industry coverage

Industry coverage	2007	2012	2017
All 43 policy sectors	0.268	0.362	0.398
Manufacturing sectors	0.288	0.380	0.489

G Robustness of Priorities and Policy Results

G.1 Sensitivity of revealed priorities to calibration

I assess how revealed priorities depend on scale elasticities, production and final-expenditure shares, and foreign-trade elasticities. I first assign a scale elasticity of 0.17 to industries with zero baseline elasticities, retaining the positive estimates. A second exercise replaces the fixed 2007 production and final-expenditure shares with those from each benchmark year's national input-output table. Further exercises set the foreign-demand elasticity to 0.73 or 1.39, or the foreign-supply elasticity to 10, compared with unit-elastic demand and perfectly elastic supply in the baseline. The foreign-demand elasticity applies to demand for the Chinese sectoral composite. The foreign-supply elasticity applies to total Chinese imports in each industry, so the resulting foreign price change is common across provincial destinations. The demand values are the endpoints of the 90 percent bootstrap confidence interval in Fajgelbaum, Goldberg, Kennedy, and Khandelwal (2020), whose estimates also inform the calibration of Adão, Becko, Costinot, and Donaldson (2026).

Each exercise repeats the priority recovery while retaining observed production and subsidies and fixed nominal provincial trade imbalances. Table G.1 reports Pearson correlations with baseline priorities across the same province–industry cells. Revealed priorities remain highly correlated with the baseline estimates under each alternative calibration.

Table G.1: Correlations of revealed priorities with the baseline

	2007	2012	2017
All 43 policy sectors			
Positive scale elasticities	0.9988	0.9987	0.9992
Annual production and expenditure shares	1.0000	0.9771	0.9610
Foreign-demand elasticity: 0.73	0.9995	0.9992	0.9992
Foreign-demand elasticity: 1.39	0.9988	0.9983	0.9983
Foreign-supply elasticity: 10	0.9999	0.9999	0.9998
Manufacturing			
Positive scale elasticities	0.9999	0.9998	0.9999
Annual production and expenditure shares	1.0000	0.9952	0.9940
Foreign-demand elasticity: 0.73	0.9996	0.9992	0.9990
Foreign-demand elasticity: 1.39	0.9990	0.9980	0.9978
Foreign-supply elasticity: 10	0.9999	0.9999	0.9999

G.2 An equilibrium with zero trade deficits

The main specification holds each province's observed nominal trade imbalance fixed. This exercise instead constructs a zero-deficit economy from that benchmark, following the equilibrium deficit-elimination approach of Caliendo and Parro (2015). I set $D_o = 0$ for every province while retaining subsidy rates, productivity and trade-cost fundamentals, factor endowments, and the foreign-trade schedules. Prices, production, expenditure, trade shares, and national productivity adjust to the resulting competitive equilibrium. Aggregate trade is balanced across each province's domestic and foreign partners, without imposing bilateral or industry-level balance.

The resulting allocation becomes the benchmark for a new priority recovery and policy counterfactuals. Rebaselining updates the shares in national input pools using physical composite inputs, preserving the scale technology. This construction differs from imposing zero deficits while holding observed production fixed and adjusting trade flows by an accounting reconciliation. The direct zero-deficit solution and a gradual elimination of deficits reach the same equilibrium; reversing the change restores the original fixed-imbalance equilibrium.

Table G.2 compares inland–coastal priority alignment. The same 3,859 province–industry cells are retained, and all 90 province-year inverse systems have full rank and negative-definite own-policy Hessians. Priority alignment increases in both periods under both specifications. The association with development plans also remains positive: in the joint provincial-plan regression, a one-standard-deviation higher priority is associated with a 3.92 percentage point higher probability of designation. Prior-growth associations remain positive and statistically significant in all three years. These comparisons repeat the recovery at unchanged subsidy rates; they do not hold the original government objectives fixed during the elimination of deficits.

Table G.2: Priority alignment after eliminating trade deficits in equilibrium

	2007	2012	2017
All 43 policy sectors			
Fixed imbalances	0.268	0.362	0.398
Zero-deficit equilibrium	0.265	0.369	0.396
Manufacturing			
Fixed imbalances	0.288	0.380	0.489
Zero-deficit equilibrium	0.285	0.377	0.484

Table G.3 reports percentage changes in national welfare relative to each economy's own benchmark, with national welfare evaluated using that benchmark's fixed provincial

factor-income weights. Subsidy removal and priority removal lower national welfare in all three years, while redesign within provincial real budgets raises it. In 2012, the redesign raises national welfare by 1.139 percent in the constructed zero-deficit economy, compared with 1.276 percent in the fixed-imbalance economy. All 30 provinces gain in the former, compared with 28 in the latter. The national welfare conclusion therefore survives, while the distribution of gains depends on trade imbalances.

Eliminating deficits and surpluses changes access to resources as well as production. The welfare change during that construction is consequently not a pure efficiency gain from balanced trade and is excluded from the policy comparisons in the table.

Table G.3: Policy counterfactuals in the fixed-imbalance and constructed zero-deficit economies

	2007	2012	2017
Remove subsidies			
Fixed imbalances	−0.390	−0.225	−0.250
Zero-deficit equilibrium	−0.385	−0.244	−0.256
Remove sectoral priorities			
Fixed imbalances	−0.270	−0.068	−0.117
Zero-deficit equilibrium	−0.319	−0.095	−0.151
Redesign within provincial real budgets			
Fixed imbalances	1.747	1.276	1.600
Zero-deficit equilibrium	1.472	1.139	1.472

G.3 Zero subsidies as inequality conditions

The main recovery treats observed signed rates, including zero rates, as interior policy choices. As a sensitivity exercise, I instead impose the nonpositive objective derivative required at a zero lower bound. One variant treats all zero rates as inequality moments; a second changes only observed zero rates, retaining equality moments for imputed zeros. Negative observed rates remain equality moments. Accounts, subsidy observations, parameters, and equilibrium derivatives are otherwise unchanged.

These inequalities generally admit multiple priority vectors. I select the vector with the smallest squared Euclidean norm subject to the retained first-order conditions. All 180 province-year-variant quadratic programs satisfy their feasibility and optimality checks. Table G.4 shows that increasing inland–coastal alignment survives both selections. The manufacturing increase is somewhat smaller when all zero rates are treated as inequalities. At the selected priorities, all 180 cases also satisfy the provincial first-order and strict local second-order sufficient conditions under their respective policy restrictions: 30 of 30 provinces

in each year and variant. These results concern the selected priority vectors, not the entire identified set. Policy counterfactuals have not been recomputed for these alternative weights.

Table G.4: Priority alignment under alternative treatment of zero subsidies

	2007	2012	2017
All 43 policy sectors			
Equality recovery	0.268	0.362	0.398
All zero rates as inequalities	0.267	0.362	0.394
Observed zero rates as inequalities	0.267	0.362	0.394
Manufacturing			
Equality recovery	0.288	0.380	0.489
All zero rates as inequalities	0.289	0.376	0.461
Observed zero rates as inequalities	0.289	0.376	0.463
Provincial objectives passing strict local SOC			
All zero rates as inequalities	30/30	30/30	30/30
Observed zero rates as inequalities	30/30	30/30	30/30

G.4 Subsidy removal with fixed productivity

To isolate the role of endogenous productivity in the subsidy-removal experiment, I hold productivity fixed as production adjusts while retaining the same benchmark allocation, trade imbalances, and other model parameters. Removing subsidies then raises national welfare by 0.464 percent in 2007, 0.329 percent in 2012, and 0.447 percent in 2017, with gains in 26, 20, and 20 provinces, respectively. With national scale responses active, the corresponding welfare changes are -0.390, -0.225, and -0.250 percent. This comparison concerns the competitive-equilibrium response to subsidy removal; it does not recover new government priorities or establish robustness to other scale-elasticity calibrations.

G.5 Provincial production externalities

This exercise asks whether increasing alignment of provincial priorities survives when production externalities operate within provinces. I replace the national productivity multiplier with a province–industry multiplier, calibrate its elasticity using the localization estimates of Graham (2009), and recover priorities again. The comparison retains the same observed production, subsidy rates, and reconciled trade shares in both specifications. It uses the benchmark with fixed nominal provincial trade imbalances. Trade elasticities, the Roy parameter, input-output coefficients, foreign-trade assumptions, and the set of 43 policy industries are also unchanged between the two specifications.

Specification and calibration. Let Ψ_{os}^{P} denote the provincial productivity multiplier. Using the physical input composite $F_{os} = B_{os}/u_{os}$ defined in Section 3, its exact-change representation is

$$\widehat{\Psi}_{os}^{P} = \widehat{F}_{os}^{\psi_s} = \left(\frac{\widehat{B}_{os}}{\widehat{u}_{os}}\right)^{\psi_s}. \tag{G.1}$$

Each province–industry has its own input pool. Industries assigned the same coefficient remain separate pools. Fundamental productivity is recalibrated at the benchmark so that the same observed allocation can be supported. In a unilateral policy change, rival policies are fixed, while prices, trade, factor allocation, and all provincial input pools respond through the competitive equilibrium. The change in the externality specification therefore changes the marginal incentives used to recover priorities.

I use the signed localization coefficients at the common one-kilometer radius in Table 4 of Graham (2009). These coefficients describe the association between firm productivity and own-industry employment in the surrounding neighborhood. I retain negative and statistically insignificant point estimates, and do not select the largest positive estimate across radii reported in the paper's Table 5. The radius and industry mapping were fixed before examining revealed priorities or second-order conditions. Urbanization coefficients, which describe a different externality, are not added to the localization coefficients.

The industry crosswalk uses detailed gross output from China's 2007 national input-output table. An exact industry match, or a narrower industry within a source group, receives that group's coefficient. When a model industry combines source groups, I take their output-weighted average. Missing source industries receive the weighted average of available members of their SIC subsection; if none is available, they receive the average across covered manufacturing groups. This latter average excludes imputed industries, primary industries, construction, and services. All aggregation weights remain fixed at their 2007 values across the three benchmark years.

Table G.5 reports the resulting coefficients for all 29 manufacturing industries. The ICT coefficient combines computers and communications/electronics, with computer output accounting for 34.24 percent of the aggregate. Electrical equipment combines a subsection average with a manufacturing average for household appliances. For measuring instruments, the detailed office/photographic-machinery component remains mixed; its composition is approximated using observed computer and instrument output as donor weights. Mapped source groups cover 68.71 percent of manufacturing output, subsection extrapolation accounts for 5.42 percent, manufacturing-wide extrapolation for 25.46 percent, and this mixed-component assignment for 0.41 percent. Externalities remain zero in the other 18 model industries, preserving the comparison's nonmanufacturing scope.

Table G.5: Provincial externality calibration by manufacturing industry

Model industries	ψ_s	Source or assignment
Food processing; food products; beverages	0.019000	SIC 15
Tobacco	0.019000	Subsection DA average
Textiles; apparel	−0.003000	SIC 17–18
Timber	0.008000	SIC 20
Paper	0.028000	SIC 21
Printing	−0.001000	SIC 22
Chemicals; pharmaceuticals; chemical fibers	0.030000	SIC 24
Rubber and plastics	0.026000	SIC 25
Ferrous metals; nonferrous metals; fabricated metal products	−0.020000	SIC 27–28
Transport equipment	0.039000	SIC 34–35
Electrical equipment	0.008649	Component averages
ICT equipment	0.011251	SIC 30 and 32
Measuring instruments	−0.014185	SIC 30 and 33
Leather; furniture; cultural and sporting goods; petroleum processing	0.008938	Manufacturing average
Nonmetallic minerals; general machinery; special machinery; other manufacturing; waste recycling	0.008938	Manufacturing average

This calibration requires transferring British neighborhood employment elasticities to Chinese provincial composite inputs. It also represents broader source industries using the model's finer classification and assumes fixed component weights adequately summarize their responses. These are additional calibration assumptions, not an exact mapping of geographic spillover reach. Graham's productivity measure is based on turnover, and the study acknowledges unresolved simultaneity and output-price effects. I therefore interpret the coefficients as conditional evidence for an alternative specification, including possible local diseconomies. I do not apply a value-added-share adjustment or transform the coefficients into internal returns to scale.

Revealed priorities and alignment. For each province-year, I solve the first-order conditions using the equilibrium derivatives under equation (G.1). Priorities remain signed and unrestricted. Table G.6 compares the resulting alignment with the matched national-externality specification. Both use the same 209 inland–coastal province pairs and hold each pair's common sector set fixed across years. Alignment increases under provincial externalities both across all 43 policy industries and within manufacturing. The annual correlations of manufacturing priorities across the two specifications range from 0.976 to 0.988.

I also evaluate the full own-policy Hessian of each provincial objective at the newly revealed priorities, including all cross-industry terms. All 90 province-year Hessians are negative

Table G.6: Priority alignment under national and provincial externalities

	2007	2012	2017
A. Inland–coastal cosine: all 43 policy industries			
National externalities	0.268	0.362	0.398
Provincial externalities	0.254	0.339	0.372
B. Inland–coastal cosine: manufacturing industries			
National externalities	0.288	0.380	0.489
Provincial externalities	0.254	0.336	0.443
C. Comparison and local optimality			
Priority correlation: all policy industries	0.991	0.983	0.989
Priority correlation: manufacturing	0.988	0.976	0.984
Provincial objectives passing strict local SOC	30/30	30/30	30/30

definite, and the maximum first-order residual is 6.66×10^{-16}. The largest Hessian eigenvalue across all cases is -8.14×10^{-6}. Analytical derivatives were checked against independently solved nonlinear equilibria, including positive and negative policy perturbations and an industry with a negative externality coefficient. The first-order conditions and curvature checks establish strict local best responses on the regular equilibrium branch. They do not establish global optimality.

The increasing alignment of priorities therefore survives this provincial-externality calibration. The exercise changes both the geographic scope and the elasticities of the externalities, and does not establish equivalence between the provincial and national specifications. Appendix G.6 reports the subsidy-removal, priority-removal, and priority-redesign counterfactuals under this same provincial-externality calibration.

G.6 Sensitivity of welfare and priority redesign

I repeat the policy experiments under the alternative calibrations in Appendices G.1 and G.5. The national-externality calibration remains the preferred specification. Each comparison uses its own calibrated benchmark, fixed nominal provincial trade imbalances, and fixed benchmark provincial factor-income weights. The subsidy-removal and priority-removal experiments follow Appendix D; the redesign retains each province's observed real subsidy budget as in Appendix E. The alternative calibrations change the relevant equilibrium responses and policy incentives, which are recomputed for each experiment. Tables G.7 and G.8 report percentage changes in national welfare relative to each calibration's own benchmark.

Table G.7 reports the 2012 results. Redesign within observed provincial budgets raises national welfare under every calibration, with gains ranging from 0.849 to 2.004 percent

across the six alternatives, compared with 1.276 percent in the preferred specification. The comparison between observed policies and the welfare-only equilibrium is more sensitive. Removing priorities raises welfare under annual production and expenditure shares, positive scale elasticities in industries with zero baseline elasticities, the lower foreign-demand elasticity, and provincial externalities. Under the provincial-externality calibration, removing all subsidies also raises welfare. That exercise changes both the geographic scope and the elasticities of externalities, as described in Appendix G.5.

Table G.7: Welfare sensitivity of policy counterfactuals, 2012

Calibration	Remove subsidies	Remove priorities	Redesign priorities
Preferred national externalities	−0.225	−0.068	+1.276
Annual production and expenditure shares	−0.090	+0.074	+1.114
Zero baseline elasticities set to 0.17	−0.716	+0.462	+2.004
Foreign-demand elasticity: 0.73	−0.167	+0.032	+1.373
Foreign-demand elasticity: 1.39	−0.304	−0.195	+1.180
Foreign-supply elasticity: 10	−0.191	−0.031	+1.246
Provincial externalities	+0.300	+0.335	+0.849

Table G.8 reports the removal experiments for 2007 and 2017. Subsidy removal lowers welfare in every national-externality calibration considered, while the sign of the priority-removal comparison varies with calibration and year. Under provincial externalities, both removal experiments raise welfare in all three years. Annual production and expenditure shares coincide with the preferred calibration in 2007, so the reported results are identical by construction.

Table G.8: Welfare sensitivity of removal experiments, 2007 and 2017

	2007		2017	
Calibration	Remove subsidies	Remove priorities	Remove subsidies	Remove priorities
Preferred national externalities	−0.390	−0.270	−0.250	−0.117
Annual production and expenditure shares	−0.390	−0.270	−0.134	−0.119
Zero baseline elasticities set to 0.17	−1.069	−0.071	−0.725	+0.278
Foreign-demand elasticity: 0.73	−0.323	−0.170	−0.171	+0.015
Foreign-demand elasticity: 1.39	−0.478	−0.393	−0.356	−0.278
Foreign-supply elasticity: 10	−0.326	−0.219	−0.193	−0.065
Provincial externalities	+0.422	+0.377	+0.415	+0.426

All reported welfare-only policy equilibria satisfy the provincial first-order and strict local second-order sufficient conditions in all 30 provinces. Each alternative 2012 redesign satisfies

the national first-order and constrained second-order sufficient conditions, and its supporting priority vectors satisfy the provincial first-order and full own-policy second-order conditions in all 30 provinces. These checks establish local optimality at the computed allocations. The alternative-calibration optimizations use one starting policy profile per exercise; the separate numerical restarts for the preferred specification are reported in Appendices C and E.

Table G.9 examines whether the redesign continues to shift priorities toward upstream manufacturing industries. For each calibration, I compare redesigned priorities with that calibration's revealed priorities and calculate upstreamness from its domestic production network using Appendix E. The sectoral correlation between mean priority changes and upstreamness remains positive, ranging from 0.453 to 0.622 across the alternatives. A province has a positive tilt when the covariance between its manufacturing priority changes and industry upstreamness is positive. This condition holds in 28 to 30 provinces across the alternative calibrations. These comparisons describe changes in priorities; they do not imply that upstream industries have the highest priority levels.

Table G.9: Priority redesign and manufacturing upstreamness across calibrations, 2012

	Correlation with mean priority change		Provinces with positive tilt
Calibration	Pearson	Spearman	
Preferred national externalities	0.612	0.570	30/30
Annual production and expenditure shares	0.522	0.579	30/30
Zero baseline elasticities set to 0.17	0.586	0.413	28/30
Foreign-demand elasticity: 0.73	0.622	0.561	30/30
Foreign-demand elasticity: 1.39	0.593	0.565	30/30
Foreign-supply elasticity: 10	0.616	0.565	30/30
Provincial externalities	0.453	0.216	29/30

G.7 Welfare-channel accounting

This appendix derives equation (20) and describes the finite-change accounting used in Sections 5.4 and 6.2. Fundamental productivities, trade costs, factor endowments, and nominal provincial trade imbalances are fixed. The national input-pool conversion coefficients in equation (B.6) also remain fixed. All income, receipt, expenditure, and subsidy terms in the differential are evaluated at the current equilibrium along the policy path.

Provincial welfare differential. The Roy allocation in equation (10) implies

$$d\log y_o = \sum_s \lambda_{os}\, d\log w_{os}, \qquad w_{os}L_{os} = \lambda_{os}\bar{L}_o y_o.$$

With fixed provincial population, the envelope for aggregate labor earnings is therefore $d(y_o\bar{L}_o) = \sum_s w_{os}L_{os}\, d\log w_{os}$. The fixed-factor endowment and market-clearing condition likewise give $d(r_o\bar{K}_o) = \sum_s r_oK_{os}\, d\log r_o$. These envelopes account for the endogenous allocation of both factors across industries.

Cost minimization implies

$$B_{os}\, d\log u_{os} = w_{os}L_{os}\, d\log w_{os} + r_oK_{os}\, d\log r_o + \sum_j P_{oj}M_{os,j}\, d\log P_{oj}.$$

Using $Y_o = y_o\bar{L}_o + r_o\bar{K}_o - \sum_s S_{os} + D_o$, $dD_o = 0$, and $d\log P_o^c = \sum_s \beta_s\, d\log P_{os}$, the factor-income envelopes and expenditure accounts give

$$Y_o\, d\log W_o = \sum_s B_{os}\, d\log u_{os} - \sum_s E_{os}\, d\log P_{os} - \sum_s dS_{os}. \tag{G.2}$$

Here $E_{os} = \beta_s Y_o + \sum_j \gamma_{js}B_{oj}$ includes final and intermediate purchases. Fixed nominal trade imbalances generate no separate dD_o term; their levels remain in Y_o, and their purchasing-power effects are retained through equilibrium income and prices.

Next, $S_{os} = \tau_{os}B_{os}$ and $B_{os} = u_{os}F_{os}$ imply

$$dS_{os} = B_{os}\, d\tau_{os} + S_{os}\Big(d\log u_{os} + d\log F_{os}\Big).$$

At fixed fundamental productivity, the producer-price equation gives

$$d\log p_{os} = d\log u_{os} - \frac{d\tau_{os}}{1-\tau_{os}} - \phi_s\, d\log Z_s.$$

Combining these expressions with $R_{os} = (1-\tau_{os})B_{os}$ yields

$$B_{os}\, d\log u_{os} - dS_{os} = R_{os}\Big(d\log p_{os} + \phi_s\, d\log Z_s\Big) - S_{os}\, d\log F_{os}. \tag{G.3}$$

Substituting equation (G.3) into equation (G.2) and dividing by Y_o gives the three channels in equation (20). The derivation also applies to the foreign-trade extension: sales receipts include exports, and expenditure and composite prices include imports. The terms-of-trade channel consequently includes price changes across all trading partners. The scale channel is the productivity contribution within this identity; the fixed-productivity counterfactual in Appendix G.4 instead allows all equilibrium adjustments to respond to the removal of that technology margin.

Finite changes and national aggregation. Let $t \in [0,1]$ index a policy path from the observed allocation to a counterfactual. Write $c_{ok}(t)$ for channel $k \in \{\mathrm{TT}, \mathrm{EES}, \mathrm{SUB}\}$ in equation (20),

replacing each differential $d\log x$ by $d\log x/dt$. Then $d\log \widehat{W}_o(t)/dt = \sum_k c_{ok}(t)$. With the fixed provincial weights in equation (29), define

$$\widehat{\mathcal{W}}(t) = \exp\left(\sum_o \omega_o \log \widehat{W}_o(t)\right), \qquad \widehat{\mathcal{W}}(0) = 1.$$

The contribution of channel k to the national welfare change is

$$\mathcal{C}_k = 100 \int_0^1 \widehat{\mathcal{W}}(t) \sum_o \omega_o c_{ok}(t)\, dt, \qquad \sum_k \mathcal{C}_k = 100\Big[\widehat{\mathcal{W}}(1) - 1\Big]. \tag{G.4}$$

The common factor $\widehat{\mathcal{W}}(t)$ converts the weighted log-welfare differentials into additive contributions to the finite percentage change. The reported units are ordinary percentage points. The provincial weights remain fixed at their benchmark values, while current provincial resources and all channel coefficients adjust along the path.

The main accounting path is linear in subsidy rates:

$$\tau_{os}(t) = (1-t)\tau^b_{os} + t\tau^{\text{cf}}_{os}.$$

At every integration node, I solve competitive equilibrium jointly with national productivity and evaluate the channel derivatives at that allocation. These intermediate policies are prescribed for accounting and need not solve the provincial policy game. Table G.10 reports all three experiments in each benchmark year. The sum of scale externalities and subsidy distortions is the net production-efficiency contribution; it is a subtotal of the two channels.

Table G.10: Welfare channels across benchmark years

Year	Experiment	Terms of trade	Scale externalities	Subsidy distortions	Total welfare
2007	Remove subsidies	+0.571	−1.024	+0.064	−0.390
	Remove priorities	+0.668	−0.906	−0.032	−0.270
	Redesign priorities	+0.958	+1.084	−0.295	+1.747
2012	Remove subsidies	+0.507	−0.771	+0.039	−0.225
	Remove priorities	+0.730	−0.709	−0.089	−0.068
	Redesign priorities	+0.750	+0.727	−0.201	+1.276
2017	Remove subsidies	+0.610	−0.916	+0.056	−0.250
	Remove priorities	+0.819	−0.865	−0.071	−0.117
	Redesign priorities	+0.893	+0.956	−0.250	+1.600

Finite channel contributions can depend on the integration path. For the three 2012

experiments, I also use a path linear in log net-of-subsidy wedges:

$$\log\Big[1-\tau_{os}(t)\Big] = (1-t)\log(1-\tau_{os}^{b}) + t\log(1-\tau_{os}^{\mathrm{cf}}).$$

Table G.11 reports each channel's contribution along the path linear in $\log(1-\tau)$ minus its contribution along the path linear in τ, in units of 10^{-6} percentage point. The largest absolute change in a channel is less than 0.00005 percentage point. Across the nine main paths and three alternative paths, the channel sums reproduce the endpoint welfare changes to within 1.3×10^{-11} percentage point. The maximum competitive-equilibrium residual is 1.2×10^{-13}, and the maximum scaled error in the analytical derivative check against independently solved finite differences is 3.6×10^{-10}. The accounting uses nine-node Gaussian quadrature after refinement checks. These checks verify the accounting and its numerical integration at the observed and counterfactual policy endpoints.

Table G.11: Sensitivity of welfare-channel contributions to the integration path, 2012

Experiment	Terms of trade	Scale externalities	Subsidy distortions
Remove subsidies	+1.798	−3.879	+2.081
Remove priorities	+49.912	−25.379	−24.533
Redesign priorities	+5.145	+40.012	−45.157

H Conditional Recovery of Relative Productivity

This appendix describes the procedure used to recover relative productivity in Section 4.2. The inversion combines production and trade accounts with restrictions on trade costs, export access, and factor measurement.

Trade restrictions and calibration targets. For each year, I impose a common bilateral component in domestic trade costs across all 47 industries, allowing industries to differ in their overall resistance to interprovincial trade:

$$\log \kappa_{ro,s,t} = b_{ro,t} + a_{s,t} \quad (r \neq o), \qquad b_{ro,t} = b_{or,t}, \qquad \kappa_{oo,s,t} = 1. \tag{H.1}$$

Combined off-diagonal log costs are nonnegative. The normalization $\min_s a_{s,t} = 0$ resolves the common-shift indeterminacy between the two additive components. Producer prices remain province–industry specific. This restriction allows trade in other industries on the same route to inform costs for an industry with sparse reported flows. No measured distance or travel time enters the calibration, and costs are estimated separately in each year.

External sales provide a second source of information. The fitted export shares satisfy

$$\widetilde{\pi}_{0o,s,t} = \frac{\exp[-\theta_s(\log p_{os,t} + \delta_{o,t})]}{\sum_i \exp[-\theta_s(\log p_{is,t} + \delta_{i,t})]}, \tag{H.2}$$

where $\delta_{o,t}$ is shared across industries within a province-year. Its coastal mean is normalized to zero. Its common level cancels from export shares and is absorbed by the foreign-demand shifter, so $\delta_{o,t}$ measures relative export access. The external destination also includes omitted Tibet, as in the benchmark accounts. The external flows come from the reconciled input-output data.

I fit domestic bilateral sales and provincial external sales jointly by minimizing Poisson deviance in flow levels, dividing each industry's contribution by its total producer receipts. The fitted allocation exactly matches each producer's total receipts, each destination's domestic purchases, and total external purchases in each industry. Individual bilateral flows and each producer's domestic/export split are approximated. At an interior optimum, a bilateral cost component matches the corresponding two-way flows summed across industries with weights proportional to θ_s divided by industry receipts. The sector component matches total interprovincial sales, and provincial export access matches the similarly weighted sum of exports. Parameters at a nonnegativity bound instead satisfy the corresponding inequalities. Producer prices are recovered jointly with the trade-cost components.

This approach follows the distinction between conditional inversion and exact-change analysis emphasized by Redding and Rossi-Hansberg (2017). Fan (2019) also calibrates productivity jointly with a common domestic trade-cost structure and approximates bilateral flows while matching regional production. Here, bilateral components are unrestricted across province pairs, and the common trade-cost structure applies to all 47 industries, including services, for which trade costs are finite.

Factor costs, prices, and the productivity residual. Observed manufacturing employment and real fixed assets anchor factor returns. Within each province, manufacturing labor payments divided by employment determine mean earnings; manufacturing fixed-factor payments divided by fixed assets determine the common rent. Both calculations use the same manufacturing cells present at all three benchmarks. The Roy allocation condition maps mean earnings and sectoral labor-payment shares into efficiency wages. Thus, observed assets proxy the model's fixed factor, and observed employment anchors effective labor through the maintained Roy structure.

The recovered domestic delivered prices determine a CES price index for each destination

and industry. If $d_{rs,t}$ is the retained domestic expenditure fraction, the total index is

$$P_{rs,t} = P^D_{rs,t} d^{1/\theta_s}_{rs,t}, \qquad P^D_{rs,t} = \left[\sum_{o \text{ active}} (\kappa_{ro,s,t} p_{os,t})^{-\theta_s} \right]^{-1/\theta_s}. \tag{H.3}$$

Foreign import offers are residuals that reproduce retained import expenditures. A province with zero recorded production remains an absent supplier, with productivity assigned zero. Its purchases remain in the system, and its input price is constructed from supplying provinces and imports using the same trade-cost restrictions.

Factor returns, efficiency wages, and intermediate prices determine $u_{os,t}$ through equation (5). For active producers, the pricing condition gives

$$A^{\text{eff}}_{os,t} = \frac{(1 - \tau_{os,t}) u_{os,t}}{p_{os,t}}, \qquad \log A^{\text{rel}}_{os,t} = \log A^{\text{eff}}_{os,t} - \frac{1}{|\mathcal{C}_s|} \sum_{i \in \mathcal{C}_s} \log A^{\text{eff}}_{is,t}, \tag{H.4}$$

where $\mathcal{C}_s$ contains coastal producers active at every benchmark. Effective productivity combines fundamental productivity and the scale multiplier. Under national EES, the common industry-year multiplier cancels in the relative measure. Producer prices are normalized to a coastal geometric mean of one within each industry-year; this does not impose equal provincial prices. Sector-common measurement constants likewise cancel in relative productivity. Observed subsidies enter the calculation through the factor $1 - \tau_{os,t}$.

Sample and aggregation. The comparison holds fixed 545 inland manufacturing province–industry cells across all three benchmarks, including 19 ICT cells. I calculate the geometric mean of relative productivity with equal weight on each retained cell. No price cap, productivity trimming, or target growth rate is imposed.